

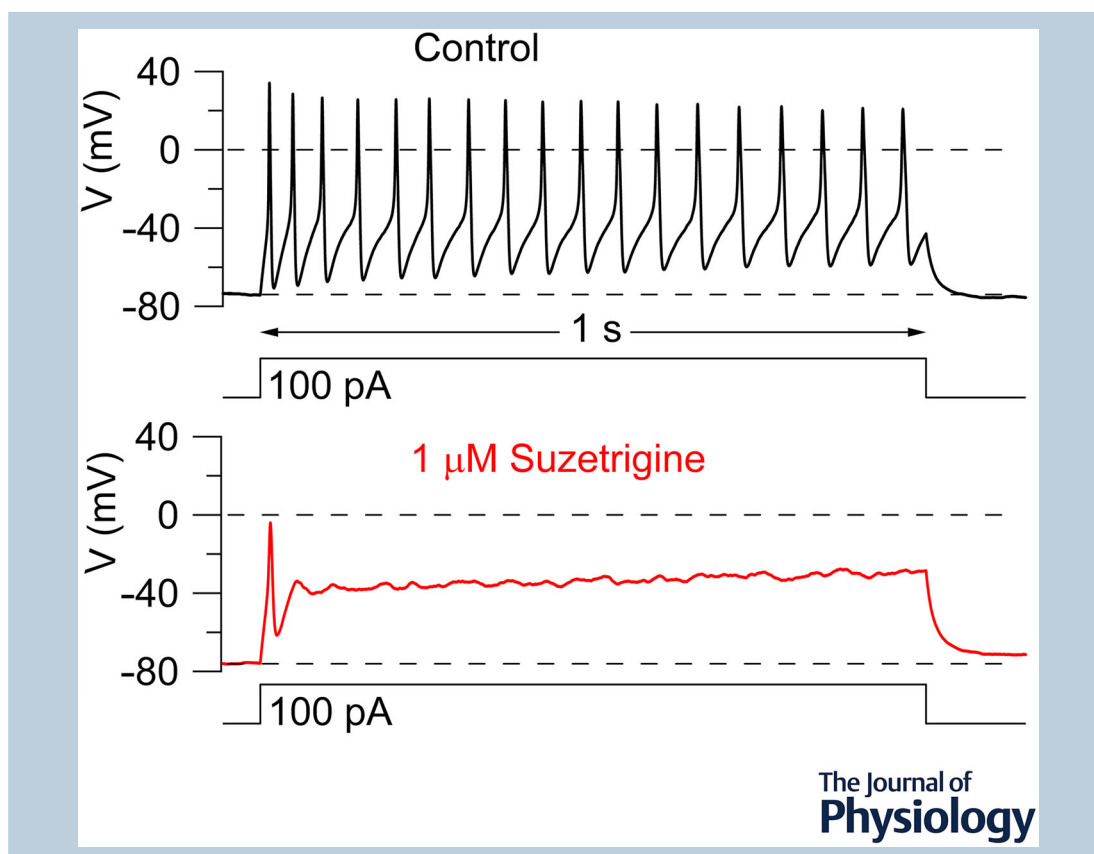
Functional role of Nav1.8 channels in action potentials of mouse CGRP-lineage dorsal root ganglion neurons

Tomás Osorno, Robert G. Stewart, Akie Fujita, Sooyeon Jo and Bruce P. Bean 

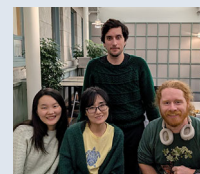
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Tomás Osorno was born in Colombia and received his BS degree from the Universidad Nacional de Colombia and PhD from Harvard University, working with Wade Regehr on cerebellar circuitry. **Akie Fujita** was born in Thailand and received BS degrees from the College of William and Mary and Columbia University and her PhD from the University of Connecticut, working with Alexander Jackson on hypothalamic circuitry. **Sooyeon Jo** was born in South Korea and received her BA from Chonnam National University and her PhD from the Gwangju Institute of Science and Technology, working with Chul-Seung Park on BK channel-interacting proteins. **Robert Stewart** was born in the USA and received his BS degree from the University of Washington and his PhD from the University of California at Davis, working with Jon Sack on Kv2 channel biophysics. They worked on this project as a team, designing experiments together and sharing cell preparations and solutions.



T. Osorno, R. G. Stewart, A. Fujita and S. Jo contributed equally.

Abstract figure legend Effect of the Nav1.8 inhibitor suzetrigine on action potential firing evoked by current injection in a mouse dorsal root ganglion neuron, showing the typical effect to reduce the height of the first action potential and inhibit repetitive firing.

Abstract Primary sensory neurons that sense noxious stimuli (nociceptors) are unusual in possessing two major components of voltage-dependent sodium channel current with different voltage dependence and kinetics. We used pharmacology to characterize the roles of defined sodium channel types in generating action potentials and controlling repetitive firing in putative peptidergic nociceptors, using mice in which CGRP-lineage neurons are fluorescently labelled. We found that the major components of sodium current were carried by Nav1.8 channels, identified by sensitivity to VX-548 (suzetrigine), and by Nav1.7 channels, identified by sensitivity to the toxin GsAF-1, with an additional smaller component of current carried by tetrodotoxin-sensitive channels resistant to GsAF-1. Action potential clamp experiments were used to quantify the size and time course of the Nav1.8 component of sodium current during firing of single and repetitive action potentials. In most neurons, Nav1.8 current carried the majority of overall sodium entry during action potentials and nearly all of the sodium current supporting action potentials during repetitive firing. Although total tetrodotoxin-sensitive current (mostly but not entirely carried by Nav1.7 channels) activated with smaller depolarizations and much more rapidly than Nav1.8 current, Nav1.8 current was equally important in setting the action potential threshold and was dominant in the action potential upstroke because of its greater size. By comparison with similar experiments in human dorsal root ganglion neurons, the results suggest Nav1.8 channels play a more dominant role in firing of mouse peptidergic nociceptors than in human C-type nociceptors, suggesting limitations of pharmacology of mouse nociceptors for predicting effects on human nociceptor excitability.

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Key points

- The new Nav1.8 inhibitor VX-548 (suzetrigine) was used to define the role of Nav1.8 currents in generating action potentials in mouse CGRP-lineage dorsal root ganglion neurons.
- Nav1.8 current typically carried the majority of sodium current evoked by voltage steps, with remaining current mainly carried by Nav1.7 channels (identified by sensitivity to GsAF-1) together with a small component from tetrodotoxin-sensitive channels not inhibited by GsAF-1.
- Nav1.8 current contributes to determining the action potential threshold, provides the majority of sodium current during the rising phase of the action potential and flows during the shoulder of the action potential typical of C-fibre nociceptors.
- Inhibition of Nav1.8 sodium current by VX-548 produces a counter-intuitive decrease in refractory period, probably resulting from reduction of potassium channel activation caused by a smaller peak and narrower action potential when Nav1.8 current is inhibited.

Introduction

Primary sensory neurons that sense noxious stimuli (nociceptors) differ from most other mammalian neurons in possessing an unusual component of voltage-activated sodium current that is resistant to tetrodotoxin (TTX) and activates and inactivates more slowly than typical TTX-sensitive sodium channels, which are also present in nociceptive neurons (Elliott & Elliott, 1993; Kostyuk

et al., 1981; Ogata & Tatebayashi, 1992; Roy & Narahashi, 1992; reviewed by Alles & Smith, 2021; Bennett et al., 2019; Goodwin & McMahon, 2021; Rush et al., 2007). The TTX-resistant component of sodium current is carried mainly by Nav1.8 channels, as shown by cloning and studies in Nav1.8 knock-out mice (Akopian et al., 1996; Cummins et al., 1999; Renganathan et al., 2001; Sangameswaran et al., 1996; reviewed by Bennett et al., 2019; Goodwin & McMahon, 2021; Han et al., 2016).

The TTX-sensitive current is carried primarily but not exclusively by Nav1.7 channels (Gingras et al., 2014; Xie et al., 2024). A large literature shows that both Nav1.7 and Nav1.8 channels are important for pain signalling in rodent pain models (Akopian et al., 1999; Alves-Simões et al., 2024; Bennett et al., 2019; Xie et al., 2024), and both are prominent targets for clinical drug development (reviewed by Alsouloum et al., 2020, 2025; Dib-Hajj et al., 2009; Waxman, 2023). A better understanding of the roles of the different sodium channels in controlling excitability of nociceptors should help interpret clinical results with selective inhibitors, and investigating differences between roles in rodent and human nociceptors (reviewed by Körner & Lampert, 2022) can help understand the strengths and limitations of rodent pain models for predicting clinical effects in humans.

The kinetic properties and functional roles of TTX-sensitive and TTX-resistant sodium channels in nociceptors have been extensively studied by a variety of complementary techniques, including genetic knock-out animals (Cummins et al., 1999; Renganathan et al., 2001) and detailed voltage clamp characterization (reviewed by Rush et al., 2007) including action potential clamp (Blair & Bean, 2002; 2003), dynamic clamp (Vasylyev et al. 2024; 2025) and computer modelling (Köster et al., 2025; Vasylyev et al., 2024, 2025; Xie et al., 2024). Selective pharmacological inhibition is particularly useful because it allows a cell-by-cell quantification of the size and kinetics of defined components of current and their roles in controlling excitability. Recently, a number of new Nav1.8 inhibitors have been introduced, including VX-548, an exceptionally potent and selective Nav1.8 inhibitor (Jones et al., 2023; Osteen et al., 2025) which was successful in clinical trials (Bertoch et al., 2025; McCoun et al., 2025) and was recently approved by the US Food and Drug Administration (FDA) for treating acute pain. Even though VX-548 is much less potent in inhibiting mouse Nav1.8 channels than human Nav1.8 channels (Ali et al., 2025; Osteen et al., 2025), it inhibits mouse Nav1.8 channels at sub-micromolar concentrations and is effective in multiple mouse models of inflammatory and neuropathic pain (Ali et al., 2025). Here, we characterized effects of VX-548 to reduce excitability in a population of putative mouse peptidergic nociceptor neurons and tested its utility to help define the functional roles of Nav1.8 channels in excitability of mouse neurons. We find that although VX-548 is imperfect as a selective inhibitor of Nav1.8 channels in mouse neurons, especially because it produces significant inhibition of voltage-dependent potassium channels at concentrations necessary for complete inhibition of Nav1.8 channels, it can be used effectively for isolating components of sodium current under voltage clamp conditions, including action potential clamp experiments characterizing the time course of Nav1.8 current flow

during action potentials. The combined results of current clamp and voltage clamp experiments show that in this population of neurons, Nav1.8 current usually contributes the majority of current supporting the upstroke of the action potential and is particularly important for enabling repetitive firing.

Methods

Ethical approval

Animal husbandry and experiments were carried out in accordance with the Guide for the Care and Use of Laboratory Animals as adopted and promulgated by the US National Institutes of Health and were approved by the Harvard Medical School Institutional Animal Care and Use Committee (Protocol IS00001369-9). The authors understood and conformed to the ethical ARRIVE 2.0 guidelines for animal use in research.

Animals

Mice were maintained as colonies of a mouse line in which the CGRP promoter drives expression of enhanced green fluorescent protein to label peptidergic nociceptors (Calca-EGFP BAC transgenic mice; Gong et al., 2003). Mice were kept in a veterinarian-supervised animal facility maintained at 21°C with four or fewer animals per cage under a 12:12 h light/dark photocycle with *ad libitum* access to food and water.

Preparation of DRG neurons

Before dissection of dorsal root ganglia (DRG), mice were deeply anaesthetized with isoflurane, placing mice in a 1 L beaker together with a gauze pad to which 200 µL of 100% isoflurane was applied, with mice separated from the gauze pad by a perforated plastic surface to avoid direct contact. After deep anaesthesia was verified by complete lack of response to a tail pinch, animals were killed by decapitation. DRG were dissected from P11–P42 mice, both male and female. Experiments used 145 mice (81 male, 65 female). The figure legends give ages and sexes of the mice used for each data set. To dissociate neurons, roots were trimmed close to the ganglia and the ganglia were treated for 15 min at 37°C with 1 mg/mL Collagenase P (Sigma #11213857001) and 50 U/mL of DNase I (Sigma, St Louis, MO, USA) in a Hanks' balanced salt solution (Gibco #14175-079) with 0.5 mM CaCl₂ and 2.5 mM MgCl₂ added. Ganglia were then treated for 2.5 min at 37°C with 0.05% trypsin EDTA (Gibco #25300054). Cells were dispersed by trituration with a P200 pipette in a Neurobasal culture medium (Gibco) supplemented with 5% FBS, 1% pen/strep and B27 supplement (Gibco

#A35828-01) with osmolarity adjusted to 310 mOsm/kg. Cells were plated in Neurobasal medium on 12 mm coverslips then incubated at 37°C (95% O₂, 5% CO₂) for 1–2 hours so that cells attached to the coverslip, after which the Petri dish was flooded with Neurobasal medium. Cells were then kept at 25 or 30°C and were used within 48 hours.

Recordings were made from fluorescently labelled neurons. At the end of most recordings, 1 μ M capsaicin was applied to the cell and almost all cells tested (132 of 135) showed TRPV1 current activated by capsaicin, and all cells expressed Nav1.8 channels based on clear responses to VX-548. Based on the co-expression of TRPV1 and Nav1.8 channels, together with the CGRP lineage of the fluorescently labelled neurons, it seems likely that most neurons correspond to a population of peptidergic nociceptors (Zheng et al., 2019). All data in Figs 3, 7, 8, 9, 10, 11 and 12 are from cells verified to be capsaicin-sensitive.

Electrophysiology

Whole-cell patch-clamp recordings were made with either an Axon Instruments Multiclamp 700B amplifier (Molecular Devices, Sunnyvale, CA, USA) controlled by pClamp9.2 software (Axon Instruments, Union City, CA, USA), filtered at 5 kHz with a low-pass Bessel filter, and digitized at 50 kHz by a Digidata 1322A data acquisition interface or a Sutter Instruments dPatch integrated amplifier/data acquisition system, with filtering at 10 kHz and digitization at 100 kHz. Whole-cell recordings were obtained using patch pipettes with resistances of 1.5–2.2 M Ω . The shank of the electrode was wrapped with Parafilm (American National Can Company) in order to reduce capacitance and allow optimal series resistance compensation (60–75%) without oscillation. Seals were obtained and the whole-cell configuration established with cells in room-temperature (21°C) Tyrode solution consisting of 155 mM NaCl, 3.5 mM KCl, 1.5 mM CaCl₂, 1 mM MgCl₂, 10 mM HEPES and 10 mM glucose, with pH adjusted to 7.4 with NaOH. After establishing whole-cell recording, cells were lifted off the bottom of the recording chamber and placed in front of an array of quartz flow pipes (250 μ m internal diameter, 350 μ m external, Polymicro Technologies) attached with styrene butadiene glue (Amazing Goop, Eclectic Products) to an aluminium rod (1 $\times$ 1 cm) whose temperature was controlled by resistive heating elements and a feedback-controlled temperature controller (TC-344B; Warner Instruments, Hamden, CT, USA). Solution changes were made (in <1 s) by moving the cell between adjacent pipes. Experiments were done with temperature controlled to either 21 or 37°C as noted in each figure.

For current clamp recordings, the internal solution was: 139.5 mM K-gluconate, 13.5 mM NaCl, 1.62 mM

MgCl₂, 9 mM HEPES, 0.99 mM EGTA, 0.09 mM CaCl₂, 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), with pH adjusted to 7.2 with KOH, with an external Tyrode solution. For the voltage clamp recordings determining dose–response for VX-548 in Fig. 1, the internal solution was: 121.5 mM CsCl, 9 mM NaCl, 1.8 mM MgCl₂, 1.8 mM EGTA, 0.18 mM CaCl₂, 14 mM creatine phosphate (tris salt), 4 mM MgATP, 0.3 mM GTP (tris salt) and 9 mM HEPES, with pH adjusted to 7.2 with CsOH; and the external solution was: 45 mM NaCl, 130 tetraethylammonium chloride (TEA Cl), 1 mM MgCl₂, 5 mM BaCl₂, 30 μ M CdCl₂, 10 mM glucose and 10 mM HEPES, with adjusted to pH 7.4 with NaOH and TEA OH, with added 300 nM TTX (for 21°C experiments) or 500 nM TTX (for 37°C experiments) in order to isolate TTX-resistant currents. The voltage clamp experiments in Figs 2 and 3 used the same K-gluconate-based internal solution as for current clamp experiments. Membrane potentials for the experiments using the K-gluconate-based solution are corrected for a liquid junction potential of –13 mV between the internal solution and the Tyrode solution in which the current was zeroed before recording.

The external solution for the experiments in Fig. 2 at 37°C had reduced sodium to improve voltage control along with TEA and Ba to inhibit potassium currents and Cd to inhibit calcium channel currents: 77.5 mM NaCl, 77.5 mM TEA Cl, 4 mM BaCl₂, 30 μ M CdCl₂, 10 mM glucose and 10 mM HEPES, pH 7.4 with NaOH and TEA OH. The external solution for the voltage clamp experiments in Fig. 3 at 21°C was: 155 mM NaCl, 3.5 mM KCl, 10 mM TEA Cl, 5 mM BaCl₂, 30 μ M CdCl₂, 10 mM glucose and 10 mM HEPES, pH 7.4 with NaOH.

For the action potential clamp experiments in Fig. 12, the internal solution was 60.75 mM CsF, 60.75 mM CsCl, 9 mM NaCl, 9 mM TEA Cl, 1.8 mM MgCl₂, 9 mM HEPES, 1.8 mM EGTA, 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with CsOH and the external solution was 155 mM NaCl, 10 mM TEA Cl, 5 mM BaCl₂, 30 μ M CdCl₂, 10 mM glucose and 10 mM HEPES, pH 7.4 with NaOH. In applying the voltage waveform as a command in the action potential clamp experiments, 8 mV was subtracted from the waveform to account for a liquid junction potential of –13 mV between the K-gluconate-based pipette solution and the bath Tyrode solution in which the pipette current was zeroed before making a seal in the original recording of the action potential and a junction potential of –5 mV between the CsF/CsCl-based internal solution of the pipette solution and the bath Tyrode solution when the pipette was zeroed in the voltage clamp experiments. Reported voltages are corrected for the liquid junction potentials. In the pooled data in Fig. 12D, data were omitted for neurons in which there was clear loss of voltage control as evidenced by large currents activating

with an irregular time course when evoked by the voltage step to -35 mV, where well-controlled currents were always a small fraction of the current evoked by the step to -25 mV.

Drugs

The synthesis of the active enantiomer of VX-548 is described in Vaelli et al. (2024). Stocks were prepared at 10 mM in DMSO and aliquoted and frozen at -20°C . GsAF-1 (Redaelli et al., 2010) (Alomone Labs, Jerusalem, Israel, Cat# STG-300) was prepared at 100 μM in purified water and aliquots were stored at -20°C . TTX stock was prepared at 3 mM in water and aliquots were stored at -20°C . All external solutions, including control solutions, contained 1 mg/mL Pluronic PF-68 (Sigma), a surfactant poloxamer without which the apparent potency of VX-548 was found to be reduced (Vaelli et al., 2024). External control solutions contained matching concentrations of DMSO as the drug-containing solutions. Solutions containing VX-548 or TTX were sonicated for 2 min while heated to 30°C .

Data analysis and statistics

Data were analysed using programs written in Igor Pro 6, 8 or 9 (Wavemetrics, Lake Oswego, OR, USA), using DataAccess (Bruxton Software) to read pClamp files into Igor Pro.

For the data analysing kinetics of currents in Fig. 2, we confined analysis to neurons in which the series resistance error (calculated by multiplying the peak current by the series resistance that remained after 70 – 80% compensation) was less than 10 mV. However, for comparing magnitudes of Nav1.8 and Nav1.7 current, we included data from all neurons, because excluding neurons with the largest currents (which tended to have the largest series resistance errors) would give misleading statistics for the size of the currents. In general, the magnitude of recorded currents measured for steps near the peak of the current–voltage relationship are reduced by series resistance errors (because the voltage reached inside the neuron is more depolarized than the command voltage and thus has a reduced driving force). Thus, the relative magnitude of Nav1.8 current (measured from the control current to that in VX-548) is probably somewhat under-estimated compared to the Nav1.7 current, because the initial control current is larger than the current after application of VX-548 and before GsAF-1.

For voltage clamp data, currents were corrected for linear capacitive and leak currents, which were determined using 5 mV hyperpolarizations delivered from the resting potential and then appropriately scaled and subtracted. For action potential clamp currents,

currents evoked by 5 ms steps from -80 to -90 mV and back were signal-averaged over 200 steps to determine leak and residual capacitive currents, which were then scaled and subtracted in a point-by-point manner from the currents evoked by the action potential waveforms.

All experiments were exploratory, and statistical results of mean $\pm$ standard deviation (SD) are descriptive and are presented to enable evaluation of reproducibility and cell-to-cell variation of parameters rather than for hypothesis-testing (Michel et al., 2020).

Results

VX-548 inhibits cloned human Nav1.8 with exceptional potency (Jones et al., 2023; Osteen et al., 2025), with a K_d of 0.07 nM at room temperature (Stewart et al., 2025). VX-548 is also exceptionally selective, requiring concentrations at least $30,000$ times higher for inhibition of Nav1.7 or other sodium channels (Jones et al., 2023). VX-548 is substantially less potent for inhibiting Nav1.8 channels in rodent sensory neurons compared to human Nav1.8 channels (Ali et al., 2025; Osteen et al., 2025) but the concentrations necessary for inhibiting Nav1.8 channels in mouse neurons are comparable to A-803467, which is also less potent on native rodent Nav1.8 channels than human Nav1.8 channels (Jarvis et al., 2007). Despite the lower potency for inhibiting mouse Nav1.8 channels relative to human Nav1.8 channels, VX-548 is effective in several mouse models of inflammatory and neuropathic pain (Ali et al., 2025). We therefore explored the use of VX-548 as a pharmacological tool to help elucidate the functional roles of Nav1.8 channels in controlling excitability of mouse DRG neurons, using a population of CGRP-lineage neurons that as a whole probably correspond to a subpopulation of peptidergic nociceptors, although recognizing that this identification is uncertain in the absence of *in vivo* characterization of neuronal function (Lawson, 2002; Lawson et al., 2002).

Figure 1A shows the dose–response for VX-548 inhibition of TTX-resistant sodium current in mouse DRG neurons, comparing effects at room temperature and at physiological temperature. The inhibition of current at room temperature could be fit assuming 1:1 binding of drug to channels with a K_d of 0.13 μM , a somewhat higher potency than previous determinations for TTX-resistant channels in mouse DRG neurons [IC_{50} values of 0.21 μM in Osteen et al. (2025) and 0.35 μM in Ali et al. (2025)]. The differences may reflect small differences in experimental protocols, for example the inclusion of the poloxamer Pluronic F-68 in the external solution in our experiments. In a previous study, we found that VX-548 inhibition of human Nav1.8 channels was temperature-sensitive, with considerably reduced potency at 37°C compared to 21°C (Stewart et al., 2025). Similarly, we found that the

potency on native mouse channels is also lower at 37°C ($K_d = 0.35 \mu\text{M}$) than at 21°C ($K_d = 0.13 \mu\text{M}$ with our methods).

We began by using VX-548 together with the Nav1.7 inhibitor GsAF-1 (Redaelli et al., 2010) to define the kinetics and relative sizes of the distinct components of sodium current carried by Nav1.8 and Nav1.7 channels in this population of CGRP-lineage neurons. Figure 2 shows the effect of 10 μM VX-548 followed by cumulative addition of 30 nM and then 100 nM GsAF-1. Testing on current evoked by a depolarizing step to -3 mV from a holding potential of -103 mV where channel availability is likely to be maximal, VX-548 rapidly inhibited a component of slowly inactivating current, leaving a much faster inactivating component of current, which was inhibited by 30 nM GsAF-1.

Figure 2C shows the currents through Nav1.8 and Nav1.7 channels defined by the sensitivity to VX-548 and GsAF-1. In this neuron, the two components of current had similar magnitude of peak current, but with very different kinetics, with the Nav1.7 component of current reaching a peak much earlier and inactivating much more rapidly than the Nav1.8 component of current. In pooled results, Nav1.7 current had a peak of $3.6 \pm 1.9 \text{ nA}$ and Nav1.8 current had a peak of $5.7 \pm 2.9 \text{ nA}$ (mean $\pm$ SD, $n = 17$). The differences in kinetics seen in the cell whose

currents are shown in Fig. 2A–C were typical. In pooled results, Nav1.7 current reached a peak in $0.28 \pm 0.07 \text{ ms}$ and Nav1.8 current in $0.80 \pm 0.19 \text{ ms}$ ($n = 10$), and Nav1.7 current decayed with a dominant time constant of $0.29 \pm 0.26 \text{ ms}$ and Nav1.8 with a dominant time constant of $1.43 \pm 0.56 \text{ ms}$ (mean $\pm$ SD, $n = 10$).

Application of 30–100 nM GsAF-1 inhibited the great majority of the fast-inactivating component of sodium current remaining after Nav1.8 current was inhibited by VX-548, consistent with this current being from Nav1.7 channels. In the neuron whose currents are shown in Fig. 2A, the current remaining in 30 and 100 nM GsAF-1 had slow kinetics, and could represent either current from Nav1.9 or a small component of Nav1.8 current remaining unblocked in 10 μM VX-548. In some neurons, there was also a sizeable component of rapidly activating and rapidly inactivating current remaining in VX-548 and GsAF-1. This current is consistent with mRNA from both Nav1.1 and Nav1.6 being seen at small but detectable levels in mouse peptidergic nociceptors (Zheng et al., 2019). Consistent with this, in another set of experiments we found a component of rapidly activating and rapidly inactivating current that remained unblocked by GsAF-1 but was inhibited by TTX. Figure 3 shows the results of experiments in which VX-548, GsAF-1 and TTX were cumulatively applied to quantify the

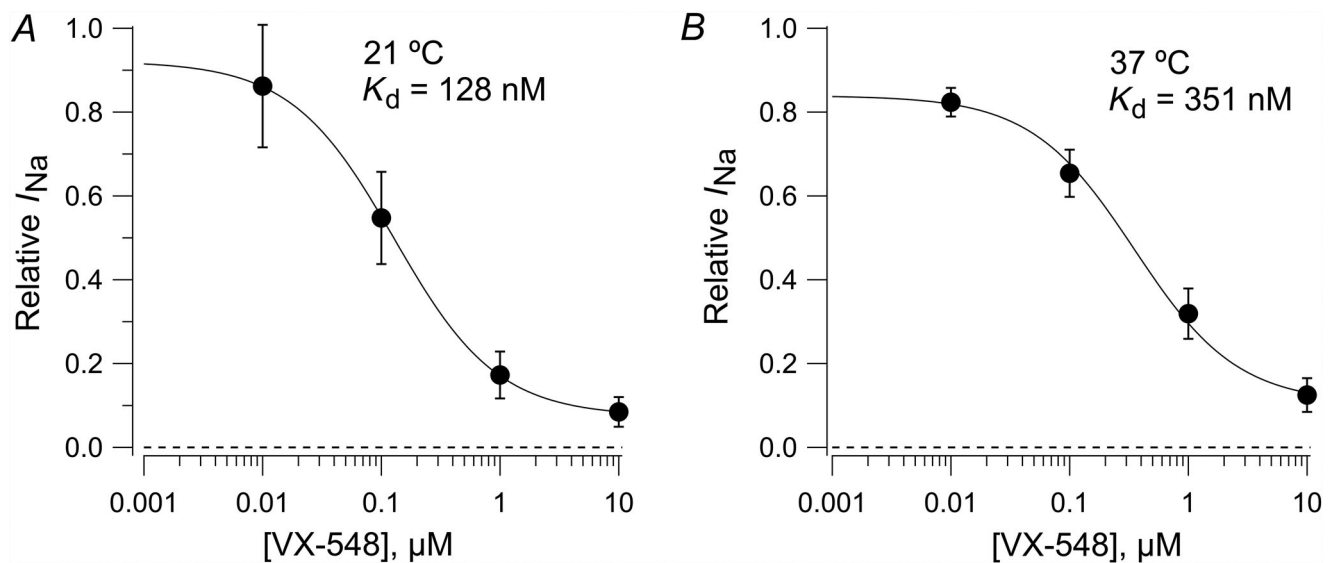


Figure 1. Dose-response for VX-548 on TTX-resistant sodium current in mouse DRG neurons

A, dose-response for inhibition of TTX-resistant current at 21°C, mean $\pm$ SD; $N = 6$ cells from two mice (P11/M/3; P13/M/3; age/sex/number of cells). B, dose-response at 37°C, mean $\pm$ SD; $N = 6$ neurons from two mice (P13/F/3; P20/F/3; age/sex/number of cells). Fitted curves: $I_{\text{min}} + (I_{\text{max}} - I_{\text{min}})/(1 + [\text{Drug}]/K_d)$, where I_{min} , I_{max} and K_d are allowed to vary and the fit included weighting of data points by SD. 21°C: $I_{\text{max}} = 0.92$, $I_{\text{min}} = 0.07$, $K_d = 128 \text{ nM}$. 37°C: $I_{\text{max}} = 0.84$, $I_{\text{min}} = 0.11$, $K_d = 351 \text{ nM}$. Currents were evoked by 30 ms steps from -95 to -5 mV . Internal: 122 mM CsCl, 9 mM NaCl, 1.8 mM MgCl_2 , 1.8 mM EGTA, 0.18 mM CaCl_2 , 14 mM creatine phosphate (tris salt), 4 mM MgATP, 0.3 mM GTP (tris salt) and 9 mM HEPES, pH adjusted to 7.2 with CsOH. External: 45 mM NaCl, 130 TEA Cl, 1 mM MgCl_2 , 5 mM BaCl_2 , 30 μM CdCl_2 , 10 mM glucose and 10 mM HEPES, pH 7.4 with NaOH and TEA OH. External solutions all contained 300 nM TTX (for 21°C experiments) or 500 nM TTX (for 37°C experiments) and 1 mg/mL Pluronic F-68.

relative size of currents from each component of current. These experiments were done with a holding potential of -73 mV, near typical values for resting potentials of the neurons, to explore the relative sizes of the currents under quasi-physiological conditions. In 16 of 17 neurons, Nav1.8 current defined by VX-548 inhibition carried the majority of the current evoked by a depolarization

to -3 mV (on average, $73 \pm 21\%$ of the summed peak currents from Nav1.8, Nav1.7 and non-Nav1.7 TTX-sensitive components, mean $\pm$ SD, $n = 17$). Of the other two components of current, Nav1.7 current defined by GsAF-1 was dominant, accounting for $19 \pm 11\%$ of the summed currents, while the non-Nav1.7 TTX-sensitive current accounted for on average $8 \pm 12\%$ of the total

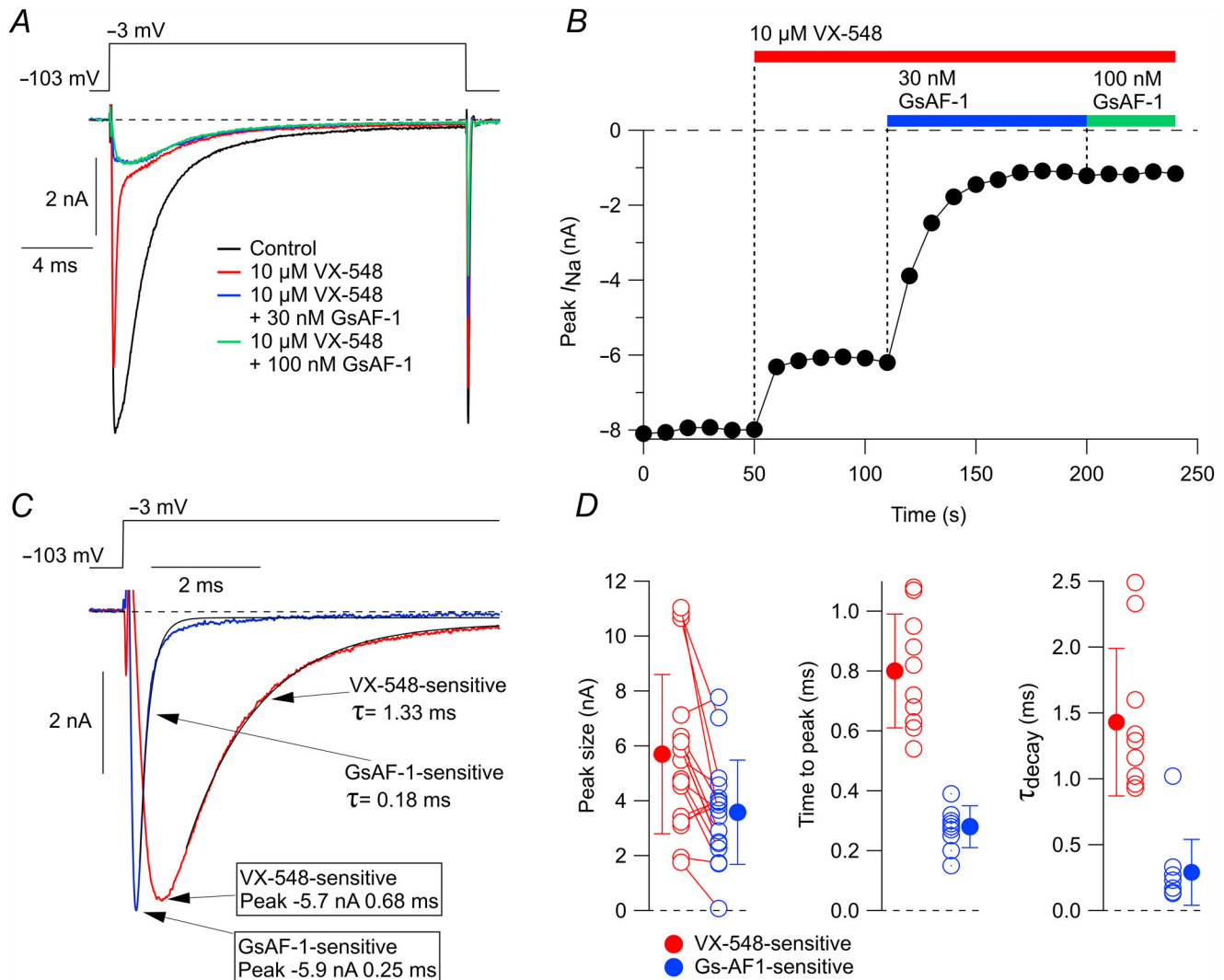


Figure 2. Definition of Nav1.8 and Nav1.7 components of sodium current by cumulative addition of VX-548 and GsAF-1 in mouse CGRP-lineage neurons

A, effect of cumulative addition of 10 μ M VX-548, 30 nM GsAF-1 and 100 nM GsAF-1 on current evoked by a step from -103 to -3 mV. **B**, time course of inhibition by VX-548 and GsAF-1 in this experiment. **C**, kinetics of Nav1.8 and Nav1.7 currents defined by inhibition by VX-548 and GsAF-1 in this experiment. Superimposed black lines are single exponential fits of the decay time constant. **D**, pooled results from application of 10 μ M VX-548 and 30 or 100 nM GsAF-1. Left, peak size of VX-548-sensitive and GsAF-1-sensitive currents. $N = 17$ neurons from P20 male mouse. Middle, time to peak for current evoked at -3 mV. Right, time constant of inactivation at -3 mV. Data for kinetic parameters are confined to 10 neurons for which series resistance error was <10 mV. Results from each neuron are shown as open circles. Filled circles show mean $\pm$ SD. Internal: 139.5 mM K-gluconate, 13.5 mM NaCl, 1.62 mM MgCl_2 , 9 mM HEPES, 0.99 mM EGTA, 0.09 mM CaCl_2 , 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with KOH. External: 77.5 mM NaCl, 77.5 TEACl, 4 mM BaCl_2 , 30 μ M CdCl_2 , 10 mM glucose and 10 mM HEPES, pH 7.4 with NaOH and TEOH. External solutions all contained 1 mg/mL Pluronic F-68. 37°C .

current and was entirely lacking or almost undetectable in 5 of 17 neurons.

We next used VX-548 to explore the consequence of selectively inhibiting Nav1.8 current on action potential firing. The effects of VX-548 on selectively inhibiting a component of overall sodium current with the expected properties of Nav1.8 current, as shown in Fig. 2, suggest a high degree of selectivity for Nav1.8 over Nav1.7 channels for the mouse isoforms of these channels, even though the potency on mouse Nav1.8 channels is much weaker than for human Nav1.8 channels. To further test selectivity of VX-548 for Nav1.8 over Nav1.7 for native mouse channels, we tested for possible inhibition of Nav1.7

current in mouse superior cervical ganglion where most current is carried by Nav1.7 channels along with smaller contributions from Nav1.3 and Nav1.6 channels (Jia et al., 2008; Ramachandra & Elmslie, 2016; Rush et al., 2006; Zhang et al., 2013) and found only small effects (8–15% inhibition by 10 μ M VX-548, $n = 3$), consistent with lack of expression of Nav1.8 channels (Rush et al., 2006) and very weak effects of VX-548 on Nav1.7 channels. In preparation for using VX-548 in current clamp experiments, we also tested for possible off-target effects on potassium channels, which would complicate interpretation of effects on action potential firing. Surprisingly, even though 10 μ M VX-548 had only small effects on Nav1.7 current,

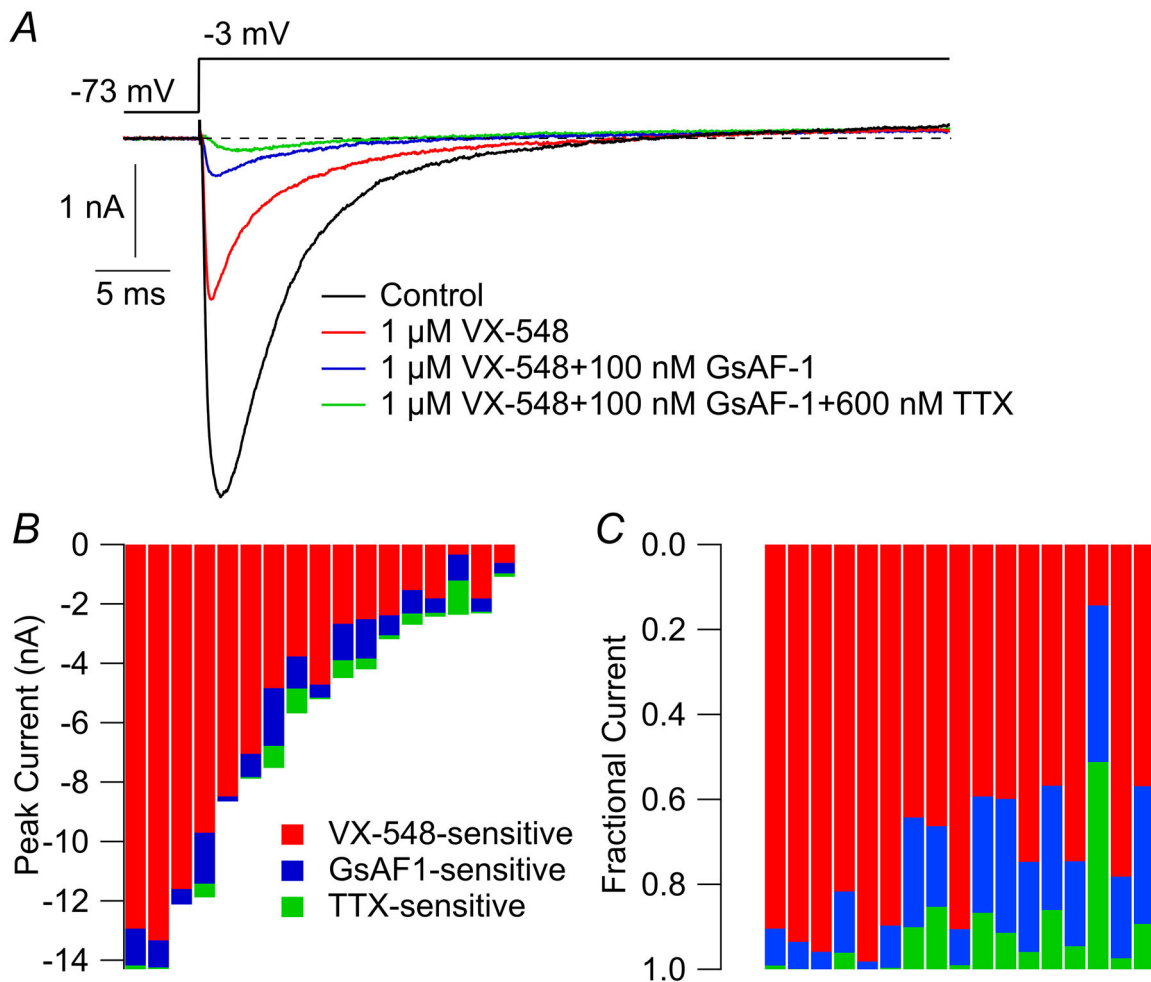


Figure 3. Quantification of components of current from Nav1.8, Nav1.7 and non-Nav1.7 TTX-sensitive sodium channels in mouse CGRP-lineage neurons with a holding potential of -73 mV

A, effect of cumulative addition of 1 μ M VX-548, 100 nM GsAF-1 and 600 nM TTX on current evoked by a step from -73 to -3 mV. B, pooled results for cumulative addition of VX-548, GsAF-1 and TTX in 17 neurons, showing the peak size of each component of current in each cell (red: VX-548-sensitive current, blue: GsAF-1-sensitive current and green: TTX-sensitive current). The bars for the neurons are arranged by size of the total current. C, comparison of fractional components of current in each cell, with same ordering of cells as in B. $N = 17$ neurons from P18 female mouse. Internal: 139.5 mM K-gluconate, 13.5 mM NaCl, 1.62 mM MgCl_2 , 9 mM HEPES, 0.99 mM EGTA, 0.09 mM CaCl_2 , 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with KOH. External: 155 mM NaCl, 3.5 mM KCl, 10 mM TEACl, 5 mM BaCl_2 , 30 μ M CdCl_2 , 10 mM glucose and 10 mM HEPES, pH 7.4 with NaOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C .

it had larger effects on potassium current, inhibiting the peak total potassium current in mouse DRG neurons evoked by a step from -73 to -13 mV by an average of $44 \pm 17\%$ (mean $\pm$ SD; $n = 5$; Fig. 4A, B). Because $10 \mu\text{M}$ VX-548 is necessary for reasonably complete inhibition of Nav1.8 current at 37°C , we therefore decided to perform current clamp experiments at room temperature (21°C) where reasonably complete inhibition of Nav1.8 current can be achieved by $1 \mu\text{M}$ VX-548. We found that VX-548 at $1 \mu\text{M}$ inhibited potassium current by $22 \pm 9\%$ at 21°C (mean $\pm$ SD; $n = 7$; Fig. 4C, D). Because the effects of inhibiting sodium current are in general expected to be opposite to those of inhibiting

potassium current, it seemed reasonable to go ahead with experiments exploring the effects of VX-548 on action potential generation and firing, using a concentration of $1 \mu\text{M}$ VX-548 and with experiments at 21°C . As will be discussed, the major effects of VX-548 were clearly from inhibition of sodium current and not from the smaller effect on potassium current.

Voltage-dependent relief of inhibition by VX-548

An additional consideration before using VX-548 in current clamp experiments on action potential firing was to test for possible voltage-dependent relief of inhibition

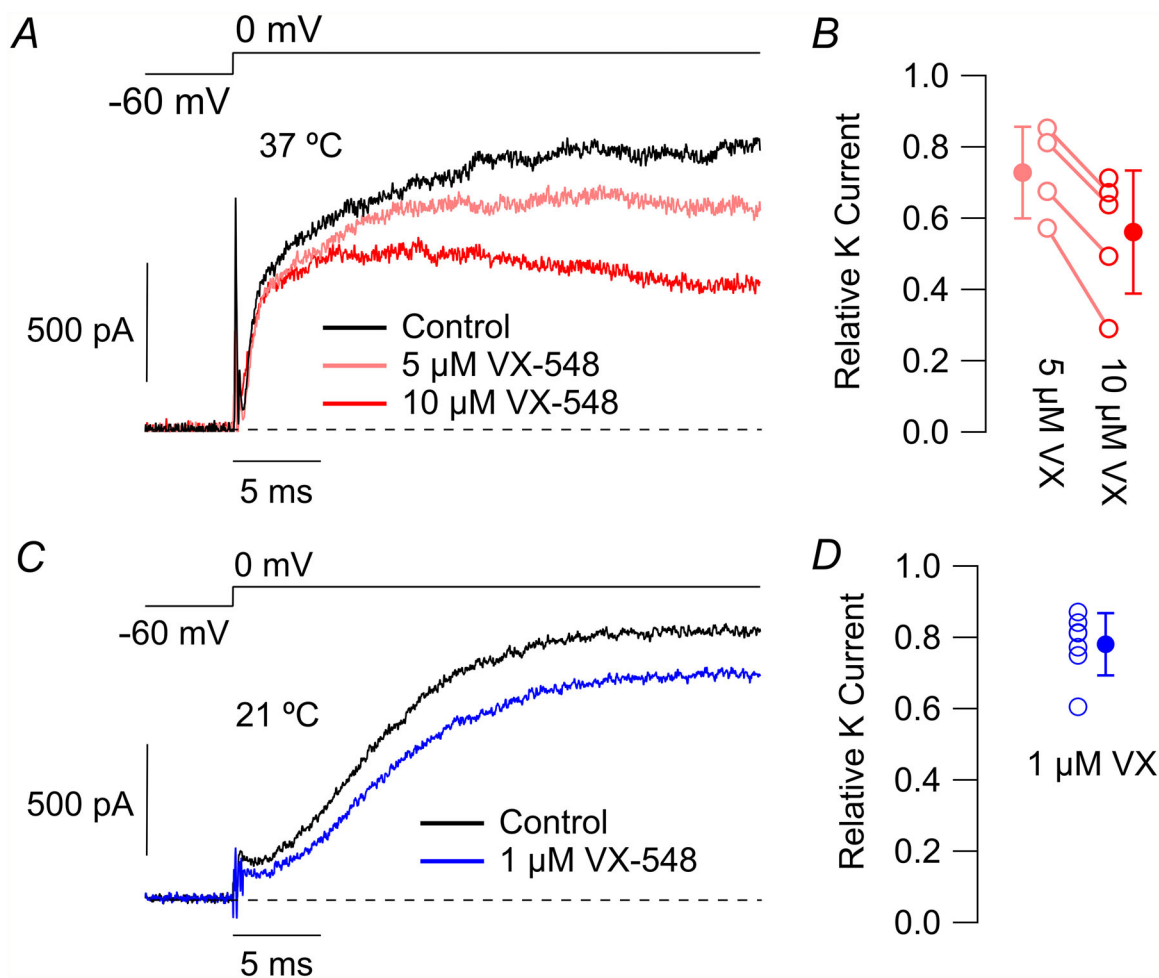


Figure 4. Effect of VX-548 on native potassium current in mouse CGRP neurons

A, effect of $5 \mu\text{M}$ followed by $10 \mu\text{M}$ VX-548 on potassium current at 37°C . B, pooled effects of $5 \mu\text{M}$ VX-548 ($n = 4$) and $10 \mu\text{M}$ VX-548 ($n = 5$) on potassium current at 37°C . Open circles show individual cells. Closed circles show mean $\pm$ standard deviation. Data from P23 male mice. C, effect of $1 \mu\text{M}$ VX-548 on potassium current at 21°C . D, pooled effects of $1 \mu\text{M}$ VX-548 on potassium current at 21°C in seven cells. Open circles show individual cells. Closed circles show mean $\pm$ standard deviation. $N = 7$ cells from two mice (P17/F/4; P34/F/3; age/sex/number of cells). Internal: 139.5 mM K-gluconate, 13.5 mM NMDG Cl, 1.62 mM MgCl_2 , 9 mM HEPES, 0.99 mM EGTA, 0.09 mM CaCl_2 , 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with KOH. External: 155 mM NMDG Cl, 3.5 mM KCl, 1.5 mM CaCl_2 , 1 mM MgCl_2 , 10 mM HEPES and 10 mM glucose, pH adjusted to 7.4 with NMDG. External solutions also contained $1 \mu\text{M}$ A-803467, $1 \mu\text{M}$ TTX and 1 mg/mL Pluronic F-68.

by physiological action potential waveforms. VX-548 is believed to inhibit human Nav1.8 channels by binding tightly to the voltage sensor region of domain 2 when it is in the non-activated, resting position, thereby stabilizing the closed state of the channel (Osteen et al., 2025). With strong depolarizations, VX-548 inhibition can be relieved, consistent with the idea that drug unbinds if the voltage sensor is forced to move to the activated, depolarized position (Jo et al., 2025; Osteen et al., 2025; Vaelli et al., 2024). In human Nav1.8 channels, the unbinding of VX-548 requires large and long depolarizations, so that there is negligible unbinding during action potential waveforms (Jo et al., 2025). However, because of the weaker potency of VX-548 on mouse Nav1.8 channels, it seemed important to test this point in native mouse Nav1.8 channels. We found that VX-548 inhibition of native mouse Nav1.8 channels can be relieved more readily and more quickly than in human Nav1.8 channels. As shown in Fig. 5A, inhibition by 1 μ M VX-548 could be partially relieved by moderate depolarizations (Fig. 5A), with an average midpoint of -22 ± 10 mV (mean $\pm$ SD, $n = 7$) for relief tested with 30 ms depolarizations. This is much more negative than for relief of inhibition in human Nav1.8 channels, which occurs with a midpoint of about $+57$ mV (Vaelli et al., 2024). In addition, the relief of inhibition of native mouse Nav1.8 channels (Fig. 5B, C) was far faster than in human Nav1.8 channels, with a time constant of 2.6 ± 1.1 ms at $+40$ mV (mean $\pm$ SD, $n = 7$) compared to a time-constant of 40 ms in human Nav1.8 channels tested at $+100$ mV (Vaelli et al., 2024). We also examined the kinetics of re-inhibition after the membrane voltage is repolarized, which occurred rapidly, with a time constant of about 3 ms (Fig. 5D).

The occurrence of faster relief at less depolarized voltages in mouse compared to human Nav1.8 channels led us to test for possible relief of inhibition during physiological action potential waveforms (Fig. 6), following methods we used previously using human Nav1.8 channels (Jo et al., 2025). We found that there is some relief of inhibition during an action potential waveform (Fig. 6A, B), presumably reflecting unbinding of VX-548 from depolarized states of the channels. However, most of the relief does not occur until late in the action potential, when the channels are mostly in the inactivated state and no longer passing current, and inhibition is re-established rapidly following the action potential, nearly as fast as the channels recover from inactivation (Fig. 6C, D). Therefore, the partial relief of VX-548 binding probably has little effect when Nav1.8 current is actually flowing during the action potential. As will be shown, this is consistent with the substantial effects of VX-548 seen in both in current clamp and in action potential clamp experiments.

In the voltage clamp experiments in Figs 2 and 3, the effects of VX-548 were rapid (steady-state in ~ 5 –10 s).

In contrast, the effects of GsAF-1 were slower (taking ~ 3 min to reach steady-state at room temperature). In preliminary current clamp experiments, we found that GsAF-1 often had only small effects, and the relatively slow inhibition made it difficult to clearly distinguish effects of GsAF-1 from changes in firing that might reflect small time-dependent changes in resting potential. Because of this, we decided to explore effects of inhibiting Nav1.7 channels by using TTX, which acts very quickly (seconds). This seemed reasonable, since the experiments summarized in Fig. 3 showed that Nav1.7 channels carry the large majority of TTX-sensitive current – recognizing, however, that the effects of TTX include not only inhibition of Nav1.7 but any other TTX-sensitive currents, which could be important in some neurons.

We first explored the effects of inhibiting TTX-sensitive and Nav1.8 channels on action potential generation when action potentials were evoked by short (0.5 ms) current injections. With this protocol, the action potential fires after the cessation of current injection, which simplifies the analysis. We first tested the effect of inhibiting the different channels on the voltage threshold for action potential firing. We found that spike threshold was sensitive to both VX-548 and TTX. Figure 7A shows a typical example of the effect of VX-548 on the threshold, determined by 0.5 ms injections straddling the level required for spike generation. VX-548 produced a clear increase in threshold from -37 to -31 mV. VX-548 produced a depolarization of spike threshold in every neuron tested (Fig. 7B), by an average of 5.0 ± 3.6 mV (mean $\pm$ SD, $n = 11$). Figure 7C shows the effect of TTX on spike threshold in a typical neuron, with a small but clear depolarization of threshold, from -29 to -26 mV. This effect was typical. In 8 of 9 neurons, spike threshold was increased by 600 nM TTX, by an average of 5.4 ± 4.4 mV (mean $\pm$ SD, $n = 9$, including a cell in which threshold was unchanged). Thus, on average, current from Nav1.8 channels and current from TTX-sensitive channels play a roughly equal role in determining action potential threshold. However, the large standard deviations suggest a substantial cell-to-cell variability in the role of each component of current.

The effects of inhibiting Nav1.8 channels compared to TTX-sensitive channels were more pronounced in later phases of the action potential. With inhibition of Nav1.8 channels (Fig. 8), there was always a decrease in action potential peak (by an average of 24 ± 12 mV, mean $\pm$ SD, $n = 11$), a decrease in maximal upstroke velocity (by an average of $54 \pm 26\%$, mean $\pm$ SD, $n = 11$) and a decrease in the action potential width (by an average of $23 \pm 10\%$, mean $\pm$ SD, $n = 11$). In contrast, application of TTX had much smaller effects on these parameters in most neurons (Fig. 9).

Figure 10 shows the results of experiments comparing the effect of TTX and VX-548 on repetitive firing of

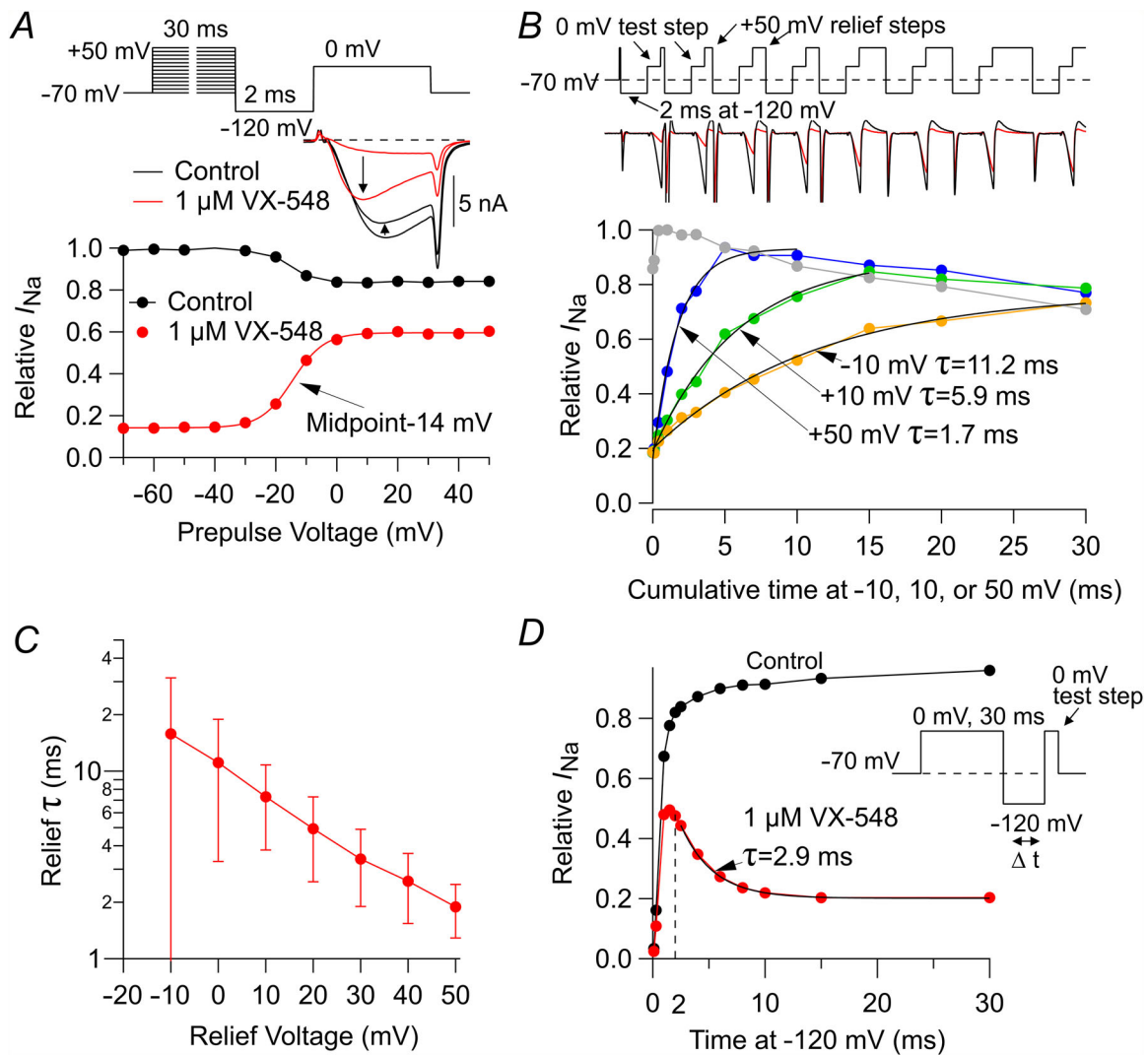


Figure 5. Voltage-dependent relief of inhibition by VX-548 of native Nav1.8 channels in mouse CGRP neurons

A, voltage-dependence of relief of VX-548 inhibition by 30 ms depolarizations. A 30 ms prepulse to varying voltages is followed by a test pulse to +0 mV to assay channel availability after a step to -120 mV for 2 ms to allow channels to recover from fast inactivation. In control, current during the test pulse decreases with increasing prepulse voltage as a result of channels entering slow inactivation. Data points in VX-548 are fit by a Boltzmann function with midpoint -14 mV and slope factor 5.2 mV. In pooled results from seven cells, the midpoint was -22 ± 10 mV (mean $\pm$ SD, $n = 7$). **B**, time course of relief of inhibition at various voltages determined with a single-sweep protocol. The fraction of drug-free channels was assayed by short (1 ms) test steps to +0 mV, each following a 2 ms period at -120 mV to allow recovery from fast inactivation; channel availability during the test steps to 0 mV is plotted as a function of cumulative time at -10, +10 or +50 mV. Data points are fitted with single exponential functions with the indicated time constants. Grey points show data recorded in control with relief steps to +50 mV. **C**, pooled data from seven cells for relief time-constant as a function of voltage (mean $\pm$ SD). **D**, time course of re-inhibition at -120 mV. After a 30 ms step to 0 mV to relieve inhibition, the voltage was stepped to -120 mV for variable times followed by a test pulse to 0 mV to assay the channel availability after re-inhibition at -120 mV. Availability as measured by peak test pulse current normalized to maximal test pulse current in control is plotted as a function of time at -120 mV. Continuous line: exponential fit with time-constant of 2.9 ms. $N = 7$ neurons from P21 male mouse. Internal: 60.75 mM CsF, 60.75 mM CsCl, 9 mM NaCl, 9 mM TEA Cl, 1.8 mM MgCl₂, 1.8 mM EGTA, 14 mM creatine phosphate (tris salt), 4 mM MgATP, 0.3 mM GTP (tris salt) and 9 mM HEPES, pH adjusted to 7.2 with CsOH. External: 77.5 mM NaCl, 77.5 mM TEA Cl, 5 mM BaCl₂, 30 μ M CdCl₂, 10 mM glucose, 10 mM HEPES and 600 nM TTX, pH 7.4 with NaOH and TBAOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C.

mouse CGRP neurons. In most neurons, VX-548 strongly inhibited repetitive firing evoked by 1 s current injections. In the example shown in Fig. 10A and B, in control the neuron fired repetitively in a graded manner with increasing magnitude of 1 s current injections, firing 20 action potentials during the maximum current injection tested (600 pA). After inhibition of Nav1.8 channels by VX-548, firing was reduced to just two action potentials regardless of the size of the current injection. As shown in pooled results in Fig. 10C, the maximum number of action potentials was usually reduced dramatically by VX-548, which reduced firing to two or fewer action potentials in

12 of 17 cells that fired repetitively in control. In contrast, inhibiting TTX-sensitive channels usually had very little effect on the ability of cells to fire repetitively during maintained current injections, with 11 of 18 neurons continuing to fire at least three or more action potentials in the presence of 600 nM TTX (Fig. 10E–G).

An interesting feature of the effects of VX-548 shown in Fig. 10A is that although VX-548 limited the firing to only two action potentials instead of many, the interspike interval between the first and second action potentials was shorter in VX-548 (Fig. 10D). A previous study in human DRG neurons showed an unexpected effect of

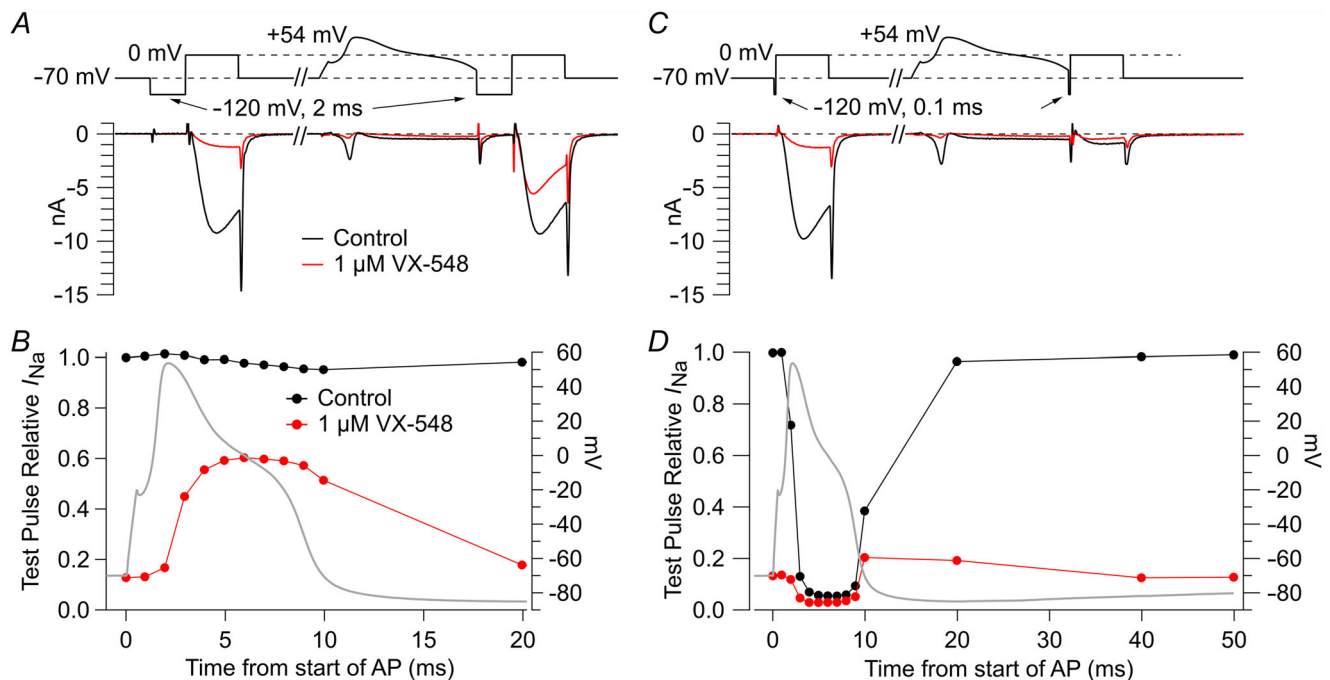


Figure 6. VX-548 inhibition is relieved by an action potential waveform but mostly after channels are inactivated

A, protocol to assay relief of VX-548 inhibition during an action potential waveform. The degree of inhibition is assayed by a 0 mV test pulse delivered after 2 ms at -120 mV to remove fast inactivation, with the sequence of -120 mV/0 mV delivered at various times after the start of the action potential waveform. The command voltage and currents are illustrated for a sweep in which the -120 mV/0 mV sequence was delivered during the falling phase of the action potential, 9 ms after the start. Relief is shown by the larger test pulse current compared to a -120 mV/0 mV sequence delivered from -70 mV without an action potential waveform. B, time course of relief from VX-548 inhibition during the action potential waveform and re-inhibition during the after-hyperpolarization. The points plot test pulse current normalized to maximum current in test pulse before the action potential waveform in control, plotted against time from start of action potential (with time zero defined as the start of current injection that evoked the action potential). Grey trace shows action potential waveform on the same time-base. C, protocol for assaying time course of channel availability, including effects of both inactivation and VX-548 inhibition, during and following the action potential waveform. The protocol is the same as in A and B except that the step to -120 mV is 0.1 ms so that channels do not recover from inactivation and the test pulse assays availability of channels that are not inactivated and not inhibited. Although VX-548 inhibition is substantially relieved during the action potential, channels are largely inactivated by the time of substantial relief. Re-inhibition after the action potential is rapid and nearly complete ~10 ms after the end of the action potential so that there is never substantial relief except when channels are mostly inactivated. Example neuron from a P21 male mouse. Internal: 60.75 mM CsF, 60.75 mM CsCl, 9 mM NaCl, 9 mM TEA chloride, 1.8 mM MgCl₂, 1.8 mM EGTA, 14 mM creatine phosphate (tris salt), 4 mM MgATP, 0.3 mM GTP (tris salt) and 9 mM HEPES, pH adjusted to 7.2 with CsOH. External: 77.5 mM NaCl, 77.5 TEACl, 5 mM BaCl₂, 30 μ M CdCl₂, 10 glucose, 10 HEPES and 600 nM TTX, pH 7.4 with NaOH and TBAOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C.

inhibiting Nav1.8 channels: a reduction in the refractory period (Stewart et al., 2025), opposite to what might be intuitively expected by reducing sodium current. We therefore explored the effect of inhibiting either Nav1.8 or TTX-sensitive channels on the refractory period (Fig. 11). The effects were very different. With application of VX-548, the refractory period was decreased in 9 of 11 neurons (Fig. 11A, B). In contrast, the refractory period was either unaffected or increased by TTX in 9 of 10 neurons (Fig. 11C, D).

To better understand the distinct roles of TTX-sensitive and Nav1.8 channels in controlling excitability of the neurons, we performed voltage clamp experiments to define the time course of Nav1.8 and TTX-sensitive currents during physiological voltage trajectories of action potential firing. First, action potential firing was recorded in current clamp using solutions mimicking physiological ionic composition. A recording of action potential firing in which a 1 s injection of current produced firing of six action potentials was used as the command waveform

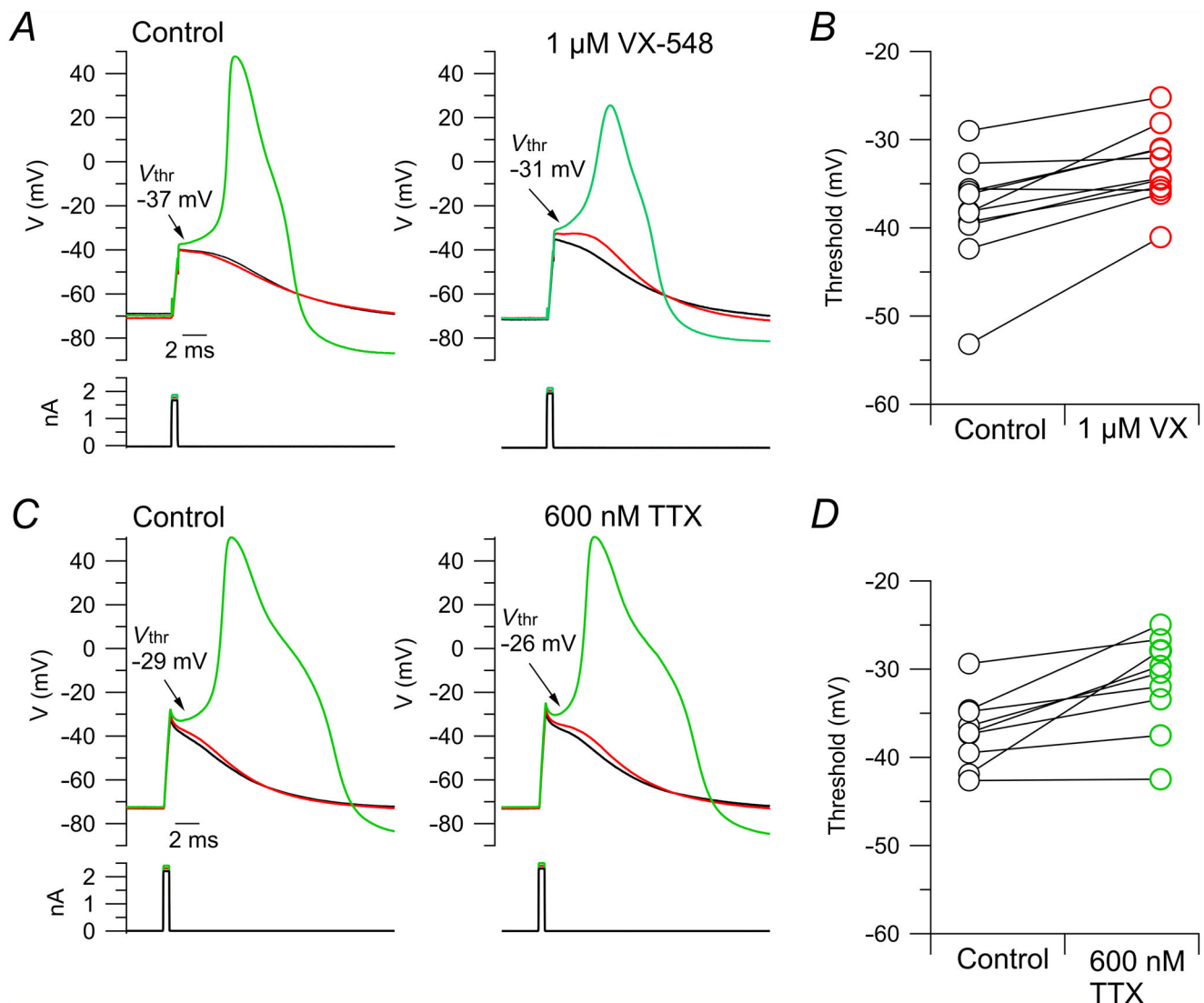


Figure 7. Effects of VX-548 and TTX on action potential threshold in mouse CGRP-lineage neurons

A, determination of action potential threshold in control (left) and after application of 1 μM VX-548 (right) in the same cell. B, pooled effects of 1 μM VX-548 on threshold. $N = 11$ cells from three mice (P17/F/2; P23/F/2; P24/M/7; age/sex/number of cells). C, determination of action potential threshold in control and after application of 600 nM TTX. D, pooled effects of 600 nM TTX. $N = 9$ cells from two mice (P24/M/5; P29/F/4; age/sex/number of cells). Internal: 139.5 mM K-gluconate, 13.5 mM NaCl, 1.62 mM MgCl_2 , 9 mM HEPES, 0.99 mM EGTA, 0.09 mM CaCl_2 , 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with KOH. External: 155 mM NaCl, 3.5 mM KCl, 1.5 mM CaCl_2 , 1 mM MgCl_2 , 10 mM HEPES and 10 mM glucose, pH adjusted to 7.4 with NaOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C.

in voltage clamp experiments (Fig. 12A). Nav1.8 current was defined by the application of 1 μ M VX-548 and total TTX-sensitive current was defined by the application of 600 nM TTX in the continuing presence of VX-548. Figure 12B shows these currents in a typical example. In the first action potential, TTX-sensitive current was smaller than VX-548-sensitive current but activated more quickly during the early rising phase of the action potential. VX-548-sensitive current activated more slowly but was larger late in the rising phase and continued to flow during the falling phase, where TTX-sensitive current had almost completely inactivated. In subsequent action potentials, TTX-sensitive current was much smaller, while VX-548-sensitive current reached

a peak somewhat larger than during the first action potential. Examination of the current components evoked by step depolarizations (Fig. 12C) help interpret the kinetics during the action potentials: TTX-sensitive current activates and inactivates much more rapidly during step depolarizations and is almost completely inactivated following a prepulse to -45 mV, while VX-548-sensitive current activates and inactivates more slowly and is still fully available after the prepulse to -45 mV. Figure 12D shows pooled results from 34 neurons studied with this protocol, all verified to be TRPV1-expressing by application of capsaicin. In 29 of the 34 neurons, VX-548-sensitive current was larger than TTX-sensitive current during the first action potential,

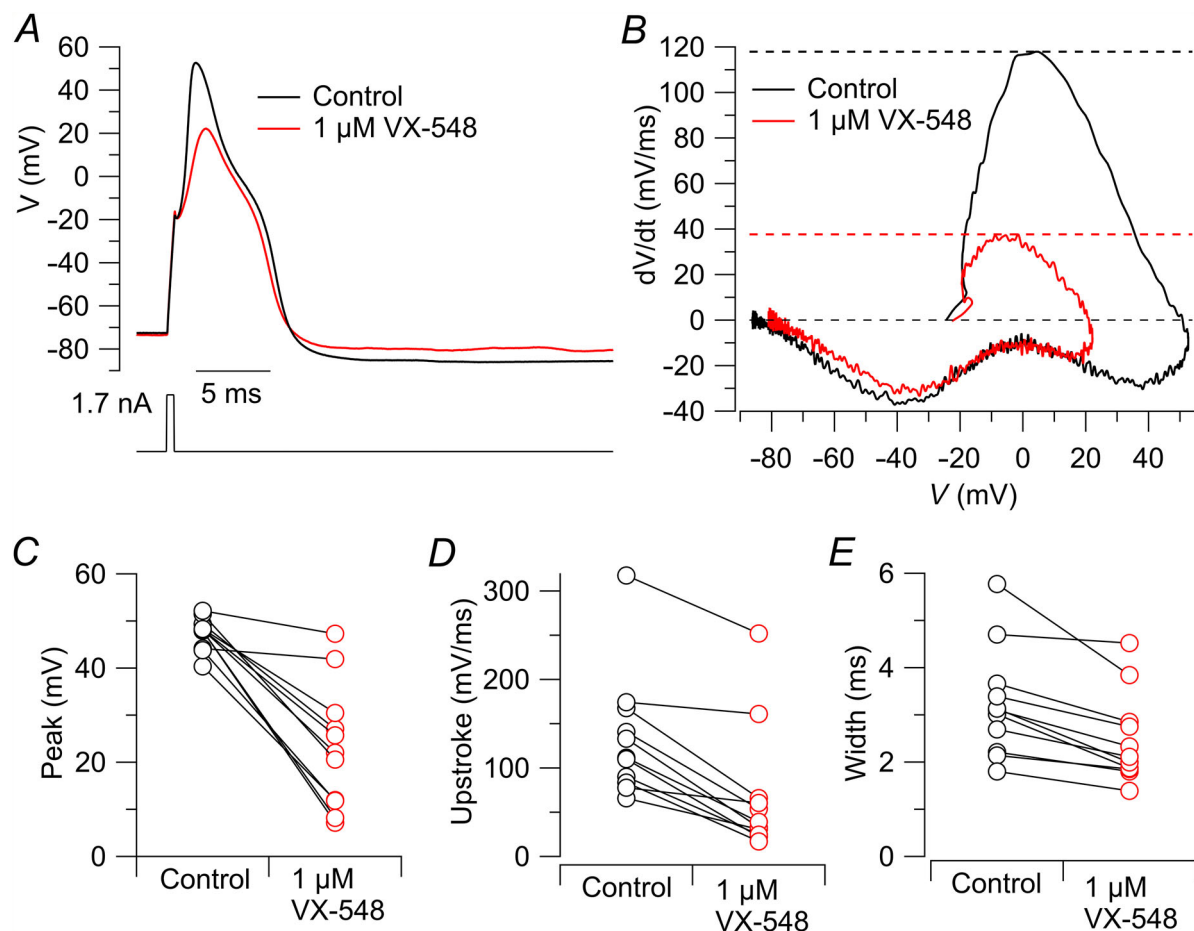


Figure 8. Effect of VX-548 on action potential shape in mouse CGRP-lineage neurons

A, action potential evoked by a short (0.5 ms) current injection before and after application of 1 μ M VX-548. B, phase-plane plot of dV/dt versus V showing dramatic reduction in upstroke velocity and peak. Upper dashed lines are drawn at the maximum upstroke velocity in control (black) and with 1 μ M VX-548 (red). C, pooled results for effect of 1 μ M VX-548 on peak of the action potential evoked by a short current injection. D, pooled results for effect of 1 μ M VX-548 on the maximal upstroke velocity of the action potential. E, pooled results for effect of 1 μ M VX-548 on the width of the action potential measured at 0 mV. $N = 11$ cells from three mice (P17/F/2; P23/F/2; P24/M/7; age/sex/number of cells). Internal: 139.5 mM K-gluconate, 13.5 mM NaCl, 1.62 mM MgCl₂, 9 mM HEPES, 0.99 mM EGTA, 0.09 mM CaCl₂, 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with KOH. External: 155 mM NaCl, 3.5 mM KCl, 1.5 mM CaCl₂, 1 mM MgCl₂, 10 mM HEPES and 10 mM glucose, pH adjusted to 7.4 with NaOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C.

often dramatically larger. Peak VX-548 current showed little change from the first to the 6th action potential, while TTX-sensitive current was much smaller during the 6th action potential than during the first.

Discussion

The results in this study add to previous studies of the role of Nav1.7 and Nav1.8 channels in the action potentials of small-diameter C-fibre cell bodies in rat and mouse DRG neurons (reviewed by Bennett et al., 2019; Goodwin & McMahon, 2021; Rush et al., 2007). In the present work, we reduced the heterogeneity of the cells studied by using a population of neurons identified by a fluorescent indicator driven by the promoter for CGRP. In addition, by using

the selective inhibitors VX-548 and GsAF-1, we were able to more precisely isolate components of currents carried by Nav1.8 and Nav1.7. The results with these inhibitors confirmed that the major component of TTX-resistant current is generated by Nav1.8 channels, consistent with earlier studies using Nav1.8 knock-out mice (Cummins et al., 1999; Renganathan et al., 2001). The results also show that the major component of TTX-sensitive current is from Nav1.7 channels, with a smaller component (on average 8% of total sodium current) from other TTX-sensitive channels. These results are consistent with results from mouse DRG neurons studied in the first several days after dissociation in another recent study using different Nav1.8 and Nav1.7 inhibitors (Xie et al., 2024).

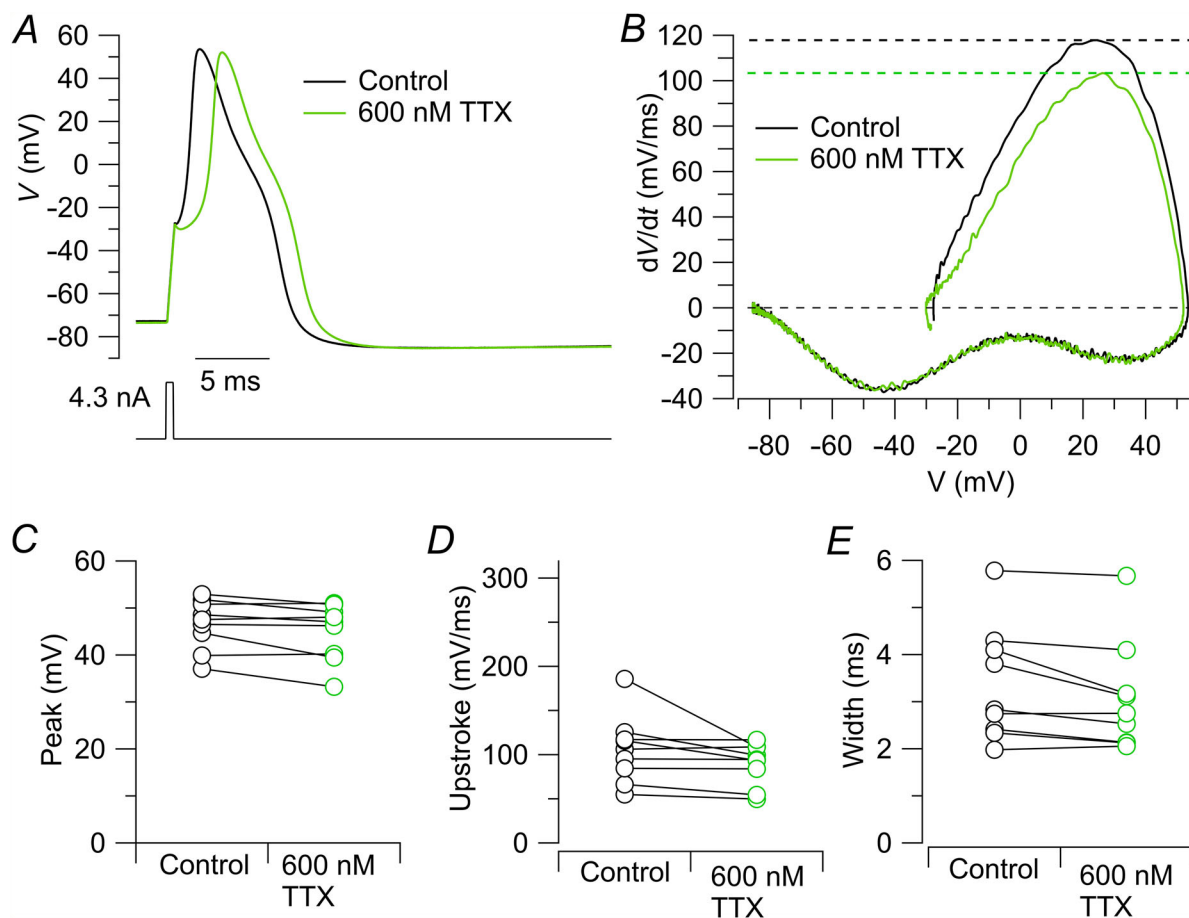


Figure 9. Effect of TTX on action potential shape in mouse CGRP-lineage neurons

A, action potential evoked by a short (0.5 ms) current injection before and after application of 600 nM TTX. B, phase-plane plot of dV/dt versus V showing minimal change in upstroke velocity or peak. Upper dashed lines are drawn at the maximum upstroke velocity in control (black) and with 600 nM TTX (green). C, pooled results for effect of 600 nM TTX on peak of the action potential evoked by a short current injection. D, pooled results for effect of 600 nM TTX on the maximal upstroke velocity of the action potential. E, pooled results for effect of 600 nM TTX on the width of the action potential measured at 0 mV. $N = 9$ cells from two mice (P24/M/5; P29/F/4; age/sex/number of cells). Internal: 139.5 mM K-gluconate, 13.5 mM NaCl, 1.62 mM $MgCl_2$, 9 mM HEPES, 0.99 mM EGTA, 0.09 mM $CaCl_2$, 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with KOH. External: 155 mM NaCl, 3.5 mM KCl, 1.5 mM $CaCl_2$, 1 mM $MgCl_2$, 10 mM HEPES and 10 mM glucose, pH adjusted to 7.4 with NaOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C.

Relative magnitude of Nav1.8 and Nav1.7 currents

Confining experiments to CGRP-lineage neurons should reduce heterogeneity of the cell population compared to previous studies in all small-diameter mouse DRG neurons. Nevertheless, a striking feature was a high degree of cell-to-cell variability in the relative sizes of the different components of sodium current, as in previous studies using all small-diameter DRG neurons (e.g. Tan et al., 2023; Thériault and Chahine, 2014; Vasylyev et al., 2024; 2025) or using neurons labelled by

a Nav1.8 promoter (Xie et al., 2024), which would also include non-peptidergic nociceptors and non-nociceptive C low-threshold mechanoreceptor neurons (Zheng et al., 2019). In our study on CGRP-lineage neurons, Nav1.8-mediated current was usually but not always (63 of 72 neurons) larger than Nav1.7 or total TTX-mediated current. In experiments done with a strongly hyperpolarized holding potential (-103 mV), peak Nav1.7 current was on average $60 \pm 36\%$ of peak Nav1.8 current (mean $\pm$ SD, $n = 17$; Fig. 2). In experiments done with a holding potential of -73 mV, which more closely

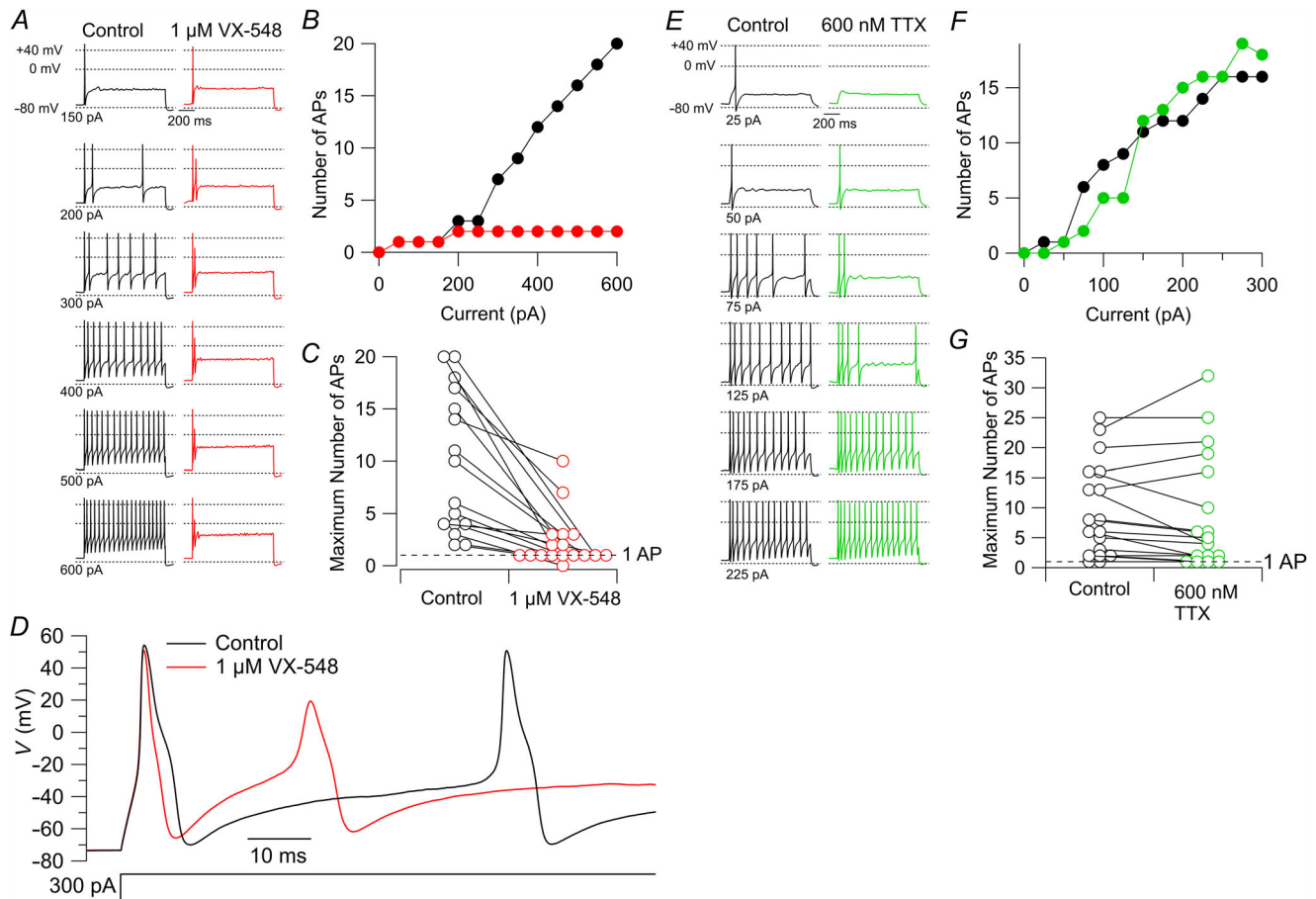


Figure 10. Effects of VX-548 and TTX on action potentials evoked by 1 s current injections

A, firing evoked by 1 s current injections of increasing magnitude before and after application of $1 \mu\text{M}$ VX-548. B, number of action potentials as a function of current injected before and after $1 \mu\text{M}$ VX-548. Action potentials were defined using criteria of action potential peak > -20 mV and height > 40 mV. To minimize effects from fluctuations in resting potential, resting potential was adjusted to near -73 mV in each condition by injection of steady current. C, pooled data from 17 neurons for maximum number of action potentials (for any current injection) before and after $1 \mu\text{M}$ VX-548. $N = 17$ cells from four mice (P12/M/4; P17/F/3; P23/F/2; P24/M/8; age/sex/number of cells). D, firing evoked by 300 pA current injection before and after VX-548 on an expanded time scale. E, firing evoked by 1 s current injections of increasing magnitude before and after application of 600 nM TTX. F, number of action potentials as a function of current injected before and after 600 nM TTX. G, pooled data from 18 neurons for maximum number of action potentials (for any current injection) before and after 600 nM TTX. $N = 18$ cells from four mice (P14/M/4; P24/M/2; P32/M/12; age/sex/number of cells). Internal: 139.5 mM K-gluconate, 13.5 mM NaCl, 1.62 mM MgCl_2 , 9 mM HEPES, 0.99 mM EGTA, 0.09 mM CaCl_2 , 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with KOH. External: 155 mM NaCl, 3.5 mM KCl, 1.5 mM CaCl_2 , 1 mM MgCl_2 , 10 mM HEPES and 10 mM glucose, pH adjusted to 7.4 with NaOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C .

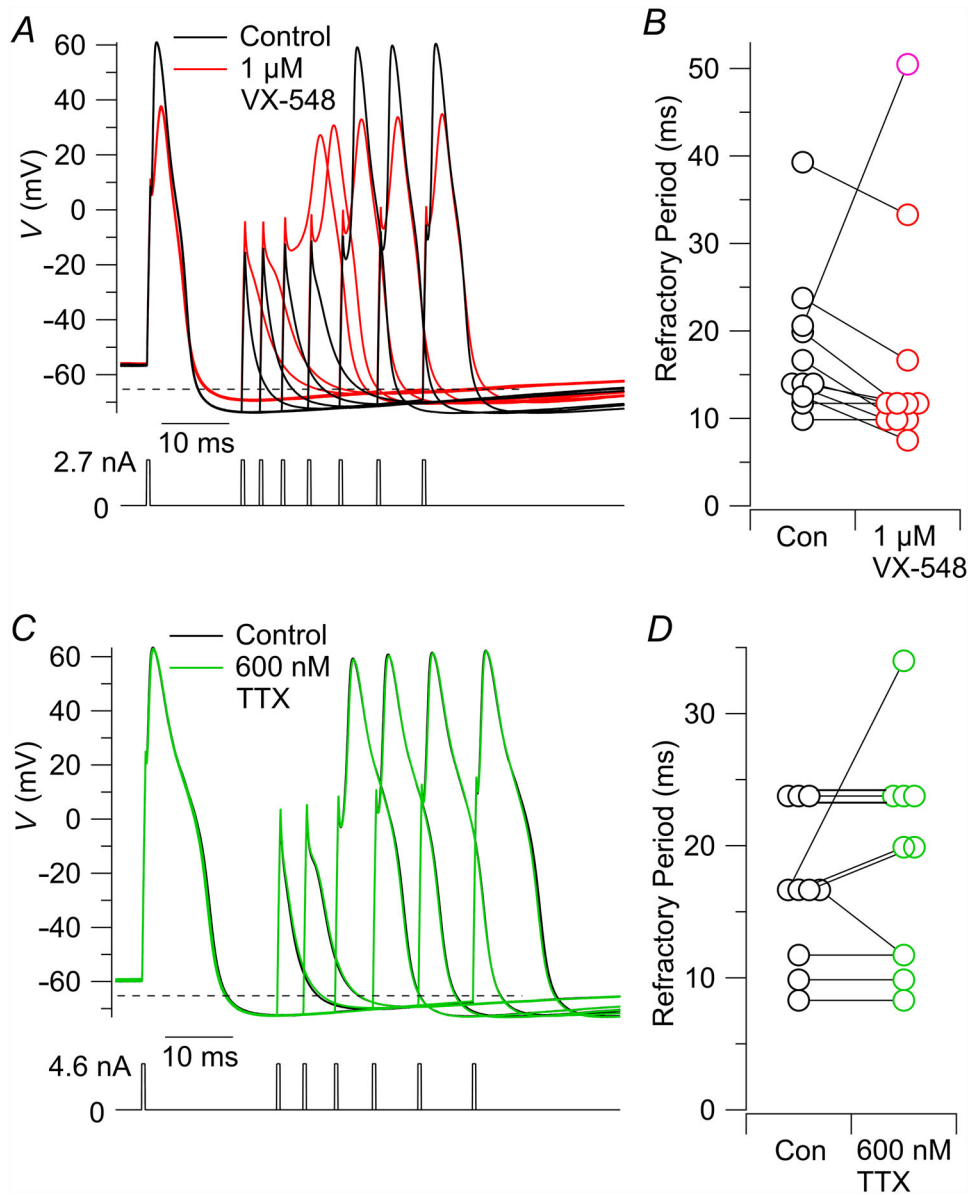


Figure 11. Differential effects of VX-548 and TTX on refractory period

A, action potentials evoked by pairs of 0.5 ms current injections with a variable time between the first and second. The stimulus magnitude of both was twice the threshold current determined in control. The time between the two current injections was varied from longer to shorter to determine the refractory period. Shown are superimposed sweeps from seven sets of times in control (black) and with 1 μ M VX-548 (red). In control, the second stimulus evoked an action potential with a spacing (start to start) of 28.4 ms but not with 23.8 ms, while after VX-548, a spacing of 19.9 ms evoked a second action potential using the same stimuli. Note reduced after-hyperpolarization in VX-548. B, pooled results, plotting the maximum spacing between the stimuli in which the second stimulus failed to evoke a spike, showing a decrease of refractory period in 9 of 11 neurons, no change in one and increase in one. $N = 11$ cells from three mice (P17/F/2; P23/M/2; P24/M/7; age/sex/number of cells). C and D, the same experiment done with 600 nM TTX, showing no effect or lengthening of the refractory period in 9 of 10 neurons. $N = 10$ cells from two mice (P24/M/6; P29/F/4; age/sex/number of cells). Internal: 139.5 mM K-gluconate, 13.5 mM NaCl, 1.62 mM MgCl₂, 9 mM HEPES, 0.99 mM EGTA, 0.09 mM CaCl₂, 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with KOH. External: 155 mM NaCl, 3.5 mM KCl, 1.5 mM CaCl₂, 1 mM MgCl₂, 10 mM HEPES and 10 mM glucose, pH adjusted to 7.4 with NaOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C.

matches the typical resting potential of DRG neurons in our recordings, Nav1.7 current was on average $40 \pm 59\%$ of peak Nav1.8 current (mean $\pm$ SD, $n = 16$), with large cell-to-cell variability (Fig. 3C). Similarly, in the action potential clamp experiments, which were done using a waveform with a natural resting potential of -73 mV, the magnitude of the peak TTX-sensitive current during the first action potential was on average $45 \pm 70\%$ of peak Nav1.8 current (mean $\pm$ SD, $n = 34$), with similarly large cell-to-cell variability (Fig. 12D). The

results show that, at least in this population of neurons defined by the promoter for CGRP, Nav1.8 current most commonly but not always plays the dominant role in action potential generation from normal resting potentials. The high cell-to-cell variability in the relative size of the components of overall sodium current may reflect distinct subpopulations of neurons, as suggested by recent transcriptomic data showing multiple distinct populations of peptidergic nociceptors (Qi et al., 2024). The new genetic tools distinguishing different sub-

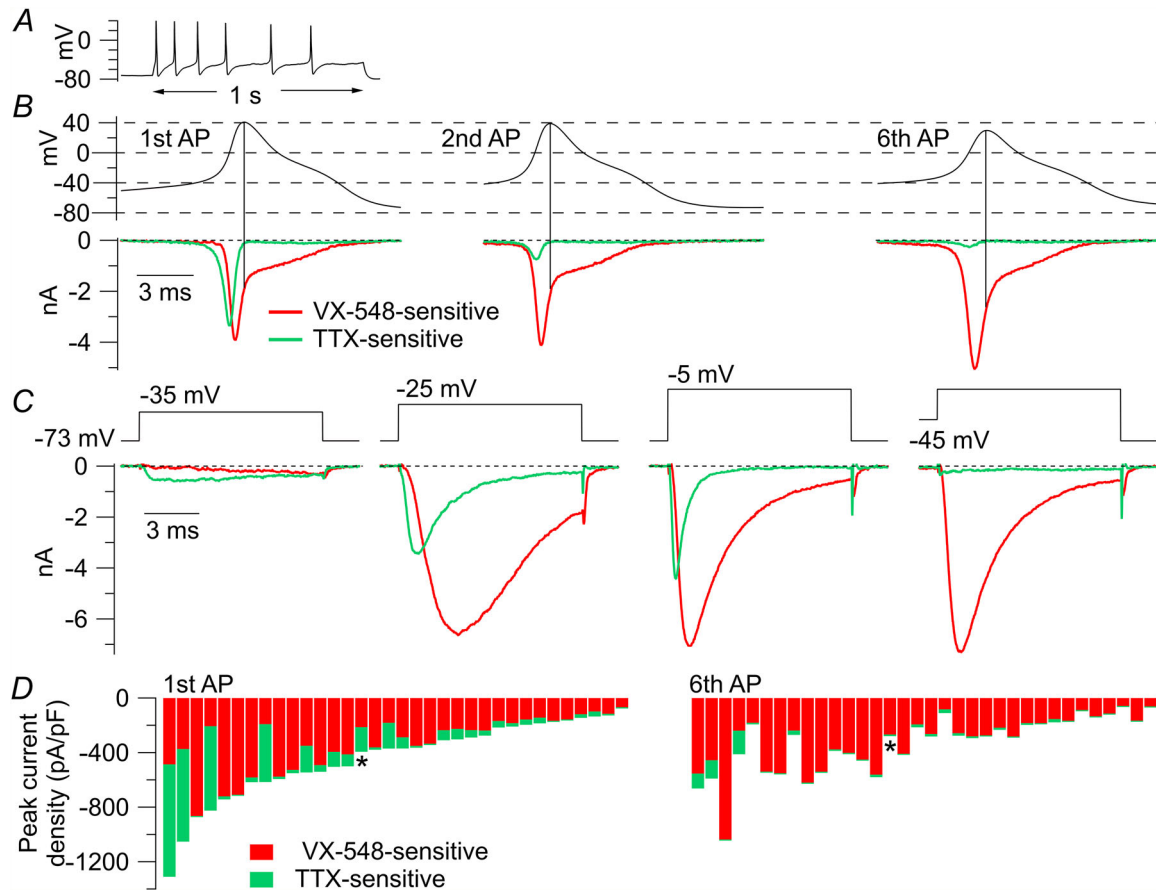


Figure 12. VX-548-sensitive and TTX-sensitive components of current during action potentials

A, record of firing evoked by 1 s current injection used as command waveform. B, currents inhibited by $1 \mu\text{M}$ VX-548 (red) and 600 nM TTX (green) during the first, second and sixth action potentials in an example. TTX-sensitive current was determined by application of 600 nM TTX in the continuing presence of VX-548. C, VX-548-sensitive and TTX-sensitive currents evoked by voltage steps (same neuron as B; voltage steps and action potential waveforms were combined in the same command voltage). Right two panels show current evoked by steps to -5 mV either directly from the holding voltage of -73 mV or after a 200 ms prepulse to -45 mV . D, pooled results in 34 neurons, showing the size of each component in each cell (red, VX-548-sensitive current; green, TTX-sensitive current) normalized to the capacitance of the cell, showing peak current densities during the first (left panel) and sixth (right panel) action potentials. Bars are arranged by size of the combined current densities in the first action potential, with same ordering for sixth action potential data. Asterisk indicates the neuron whose currents are shown in B and C. $N = 34$ cells from five mice (P29/F/7; P32/F/9; P38/M/4; P42/F/7; P42/M/7; age/sex/number of cells). Internal: 60.75 mM CsF, 60.75 mM CsCl, 9 mM NaCl, 9 mM TEA Cl, 1.8 mM MgCl_2 , 9 mM HEPES, 1.8 mM EGTA, 14 mM creatine phosphate (tris salt), 4 mM MgATP and 0.3 mM GTP (tris salt), pH adjusted to 7.2 with CsOH. External: 155 mM NaCl, 10 mM TEA chloride, 5 mM BaCl_2 , $30 \mu\text{M}$ CdCl_2 , 10 mM glucose and 10 mM HEPES, pH 7.4 with NaOH. External solutions all contained 1 mg/mL Pluronic F-68. 21°C .

populations of peptidergic nociceptors should allow future examination of the electrophysiology of these subpopulations.

A recent computational study showed that Nav1.9 channels can in principle contribute to the rising phase and shoulder of nociceptor action potentials (Köster et al., 2025). We did not explicitly test for a contribution of Nav1.9 current, but it seems likely to have been small under our recording conditions. In principle, Nav1.9-mediated current might contribute to the small currents that remained in the combined application of 1 μ M VX-548 and TTX, which typically inhibited ~90–95% of the total inward current (e.g. Fig. 3A). However, this current seemed most likely to correspond primarily to the ~15% of Nav1.8 current that would be expected to remain in 1 μ M VX-548, based on the K_d of 0.13 μ M at room temperature, and it had kinetics consistent with remaining Nav1.8 current. This is in line with previous recordings from small-diameter mouse and rat DRG neurons showing that although the neurons typically have large Nav1.9-mediated currents that can be revealed by holding the membrane at strongly hyperpolarized potentials for minutes, there is generally little or no detectable Nav1.9 current with steady holding voltages depolarized to –90 mV or so in the absence of modulation by inflammatory mediators or G-protein activation (Bosmans et al., 2011; Cummins et al., 1999; Maingret et al., 2008; Ostman et al., 2008). Evidently there is some form of ultra-slow inactivation of Nav1.9 channels that results in nearly complete loss of availability at normal resting potentials, at least with typical recording conditions in dissociated neurons.

Non-Nav1.7 TTX-sensitive sodium channels

Although GsAF-1 inhibited the majority of the current remaining after inhibition of Nav1.8 channels by VX-548, many cells had an additional component of TTX-sensitive current remaining in the presence of both VX-548 and GsAF-1 (Fig. 3). Another recent study using pharmacology to distinguish functional roles of Nav1.7 and Nav1.8 channels in mouse DRG neurons revealed the presence of a population of channels resistant to the Nav1.7 inhibitor PF-05089771 but sensitive to the Nav1.1/Nav1.3 inhibitor ICA-121431 (Xie et al., 2024). This is consistent with the detection of small but non-negligible expression of Nav1.1 mRNA in peptidergic nociceptors (Zheng et al., 2019), which also have some expression of Nav1.6 mRNA. Any of these channels could account for the component of TTX-sensitive current not blocked by GsAF-1. Although this component of current was small on average, in some neurons it was not dramatically smaller than Nav1.7-mediated current.

VX-548 as an Nav1.8 inhibitor in studies of mouse neurons

The results show that VX-548 is a useful but imperfect tool for defining the roles of Nav1.8 current in mouse DRG neurons. The concentrations of VX-548 needed for inhibiting native mouse Nav1.8 current (K_d of 0.13 μ M at room temperature) are lower than for PF-01247324, another Nav1.8 inhibitor (IC_{50} 0.53 μ M, Payne et al., 2015) that has been used to investigate the functional roles of Nav1.8 currents (Xie et al., 2024). The high degree of selectivity of VX-548 makes it ideal for distinguishing Nav1.8 current from Nav1.7 current in voltage clamp experiments where potassium currents are inhibited. However, we were surprised to find that VX-548 produces obvious inhibition of mouse neuron potassium channels even at concentrations that have almost no effect on mouse Nav1.7 channels. This reduces its utility in current clamp experiments on mouse neurons. This is especially a limitation for potential studies at physiological temperature, where a concentration of VX-548 of 10 μ M inhibited overall potassium current by ~44%. As a result, we performed current clamp experiments at room temperature, taking advantage of the lower K_d for VX-548 inhibition of Nav1.8 compared to 37°C. In the current clamp experiments, we used 1 μ M VX-548, which produced ~85% inhibition of Nav1.8 current and ~22% inhibition of potassium current at room temperature. The incomplete inhibition of Nav1.8 channels and the effects on potassium channels must be considered in interpreting the results of the effects of VX-548 in current clamp. Fortunately, however, most of the effects of VX-548 – increasing action potential threshold, reducing action potential upstroke and peak, narrowing the action potential, reducing repetitive firing – were effects that clearly reflect inhibition of sodium channels and opposite to the effects expected from inhibition of potassium channels. Thus, it seems almost certain that these effects of VX-548 – which were all dramatic – actually under-estimate what would be seen with 100% Nav1.8 inhibition without potassium channel inhibition.

A second potential limitation of VX-548 as an Nav1.8 inhibitor in mouse neurons relates to the partial relief of inhibition by depolarization. Unlike for the more potent inhibition of human Nav1.8 channels (Jo et al., 2025), we found that there was some relief of VX-548 inhibition of native mouse Nav1.8 channels during an action potential waveform (Fig. 6), because relief is much faster and requires less strong depolarizations than with human Nav1.8 channels. However, substantial relief does not occur until late in the action potential, when channels are mostly inactivated, and inhibition is re-established rapidly following the action potential (Fig. 6C, D). The results suggest that when using VX-548 to define Nav1.8

current during an action potential waveform by subtracting sweeps before and after VX-548, there is probably a slight underestimation of the magnitude of the Nav1.8 current during the shoulder of the action potential, because there is some relief of inhibition during the sweep recorded in the presence of VX-548. However, the effect is likely to be minor, consistent with sizeable VX-548-sensitive currents during the shoulder (Fig. 12) that are generally comparable in size and time course to those defined by Na replacement in the presence of TTX (Blair & Bean, 2002). Because re-inhibition of channels after an action potential is rapid and almost complete by 30 ms after the action potential, inhibition would be fully re-established between action potentials at firing frequencies of up to 30 Hz. This is consistent with the ability of VX-548 to very effectively inhibit repetitive firing of the neurons (Fig. 10), which typically reached maximal frequencies of <20 Hz. Overall, the results suggest that in mouse neurons VX-548 unbinds substantially during action potential waveforms, but because unbinding is late in the waveform where channels are mostly inactivated and re-binding occurs rapidly, the relief of inhibition does little to limit the effects of VX-548 on action potential firing, consistent with its large effects on action potential parameters and repetitive firing.

Although the voltage-dependent relief of inhibition by VX-548 must be considered in interpreting its effects, the complications from this effect are clearly much less than for a previous Nav1.8 inhibitor, A-887826 (Zhang et al., 2010), where there is much more relief of inhibition during action-potential waveforms under physiological conditions (Jo et al., 2023, 2025).

Distinct roles of TTX-sensitive and Nav1.8 current in the action potential

The combination of current clamp and action potential clamp experiments clearly shows distinct roles of TTX-sensitive and Nav1.8 currents in generating the action potential in the population of peptidergic nociceptors that we studied. Although much smaller than Nav1.8 current, TTX-sensitive current activated faster, as shown in the action potential clamp experiment in Fig. 12B, so that TTX had a substantial effect on the spike threshold (Fig. 7) even though Nav1.8 current dominated by the time of maximum upstroke of the action potential, as reflected in the larger changes in maximum upstroke produced by VX-548 (Fig. 8) relative to TTX (Fig. 9). A very dramatic difference in the kinetics of the two currents is that TTX-sensitive current was completely inactivated by the peak of the action potential, so that there was no TTX-sensitive current contributing to the shoulder of the action potential. In contrast, there was

substantial Nav1.8 current during the shoulder of the action potential, as predicted by less direct ion substitution experiments with action potential clamp (Blair & Bean, 2002). The Nav1.8 current during the shoulder probably reflects a combination of incomplete inactivation of Nav1.8 channels during the action potential along with 'resurgent' sodium current flowing during recovery of some of the inactivated Nav1.8 channels (Tan et al., 2014; Xiao et al., 2019, 2022). VX-548 inhibition of the Nav1.8 current flowing during the shoulder would contribute to the dramatic narrowing of the action potentials produced by VX-548.

Role of Nav1.8 channels in supporting repetitive firing

The results summarized in Fig. 10 showed that VX-548 had a dramatic effect to reduce repetitive firing with steady injected currents, while TTX usually did not. These results add to previous results suggesting that, under most circumstances, Nav1.8 current plays an important role in enabling repetitive firing of small DRG neurons of rat and mouse (Blair & Bean, 2003; Elliott & Elliott, 1993; Matsutomi et al., 2006; Rush et al., 1998; Harty & Waxman, 2007) because even when cells possess sizeable expression of Nav1.7 or other TTX-sensitive channels, these channels become almost completely inactivated at typical interspike voltages near -50 to -40 mV during repetitive firing and carry greatly reduced current after the first action potential (Fig. 12).

Effects of inhibiting Nav1.8 channels on refractory period

In a previous study in human DRG neurons (Stewart et al., 2025), we found that in most neurons, inhibiting Nav1.8 channels reduced the refractory period, a counter-intuitive effect of inhibiting sodium channels. In the current study, we similarly saw a reduction in refractory period with VX-548 inhibition of Nav1.8 channels in 9 of 11 mouse neurons. The effect of inhibiting Nav1.8 channels to reduce the refractory period is similar but smaller and less consistent than in human neurons. A possible mechanism for a reduction of the refractory period by inhibition of Nav1.8 channels is suggested by the observation of a reduced after-hyperpolarization by VX-548 (Figs 8, 11A), which was typical with inhibition of Nav1.8 channels but not TTX-sensitive channels (Figs 9, 11C). The reduced after-hyperpolarization produced by VX-548 is probably caused by the reduction in action potential peak and action potential width, both of which are expected to result in decreased activation of both purely voltage-activated potassium channels and also calcium-activated potassium channels. In mouse neurons,

the small (~22%) direct inhibition of voltage-dependent potassium current by 1 μ M VX-548 could also contribute to the reduced refractory period, but this would not occur with the 10 nM concentration used in human neurons, where the effects on refractory period were similar. In the case of mouse neurons, it is conceivable that the relief of VX-548 inhibition occurring during the first action potential might not be fully reversed (with re-inhibition occurring with a time constant of ~6 ms) by the time of the second stimulus, especially in cases with the shortest refractory periods in the range of 10–15 ms. However, direct measurements of the overall availability of un-blocked Nav1.8 channels during and after the action potential (Fig. 6D) show that this effect would be very minor.

The effect of Nav1.8 current to increase the refractory period suggests a mechanistic link between two characteristic features of action potential firing by C-fibre neurons (Djouhri et al., 1998, 2003), the presence of a shoulder on the falling phase (from combined calcium current and Nav1.8 current) and relatively slow firing frequencies. The broad shoulder may help produce a larger post-spike potassium conductance and stronger after-hyperpolarization, tending to increase the refractory period. It remains to be determined what potassium conductances are altered by changes in the action potential shape when Nav1.8 channels are inhibited and how this decreases refractory period even when combined with a reduction of the major component of sodium current.

Comparison with human DRG neurons

The effects of inhibiting Nav1.8 channels in this population of mouse DRG neurons can be compared to the effects of inhibiting Nav1.8 channels in human DRG neurons (Stewart et al., 2025). Overall, many effects are similar: a reduction of action potential peak, narrowing of the action potential, decrease in refractory period and inhibition of repetitive firing. The major difference is that most of the effects of inhibiting Nav1.8 channels are more prominent in the mouse neurons than in the human DRG neurons. Notably, inhibiting Nav1.8 channels had a greater effect on action potential threshold in mouse neurons than human neurons, and also a larger effect on the maximum upstroke velocity of the action potential. The comparison suggests that while both mouse and human DRG neurons have a prominent role for Nav1.8 channels in action potential shape and support of repetitive firing, there is apparently a larger relative role of non-Nav1.8 channels in human DRG neurons compared with mouse DRG neurons, at least in this population of CGRP-lineage mouse neurons.

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Additional information

Data availability statement

All data are contained in the figures.

Competing interests

None declared.

Author contributions

T.O., R.G.S., A.F. and S.J. designed and executed experiments, analysed data, and contributed to writing the manuscript; B.P.B.

helped design experiments, analyse data and contributed to writing the manuscript.

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Keywords

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Supporting information

Additional supporting information can be found online in the Supporting Information section at the end of the HTML view of the article. Supporting information files available:

Peer Review History

Authors' translational perspective

Primary nociceptive neurons express multiple voltage-activated sodium channels that are current targets for new drugs designed to inhibit pain signalling by inhibiting excitability of nociceptors. Better understanding of the functional roles of the different types of channels can help evaluate the efficacy of inhibiting each channel alone or multiple channels in combination, and a detailed understanding of their roles in mouse nociceptors will help evaluate potential limitations of mouse models of pain for predicting effects in humans.